



Chemical structure of premixed ammonia/hydrogen flames at elevated pressures

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ABSTRACT

Because it is a carbon free fuel with high volumetric and gravimetric hydrogen density, ammonia is considered to be a promising hydrogen carrier molecule; its combustion chemistry, consisting of ammonia and ammonia/hydrogen blends, are of great importance in engine and gas turbine systems. This paper presents experimental data and kinetic modeling of the structure of $\text{NH}_3/\text{H}_2/\text{O}_2/\text{Ar}$ premixed flames at elevated pressures. Equivalence ratios were maintained at 0.8, 1.0 and 1.2, and the NH_3/H_2 ratio was 1:1 (molar ratio). Experiments were performed at pressures of 4 and 6 atm. Eight recently published chemical-kinetic mechanisms of ammonia combustion and oxidation were applied to numerically simulate flame structure. Experimental and numerical data showed that the main nitrogen containing compounds in the post flame zone were N_2 and NO , while the concentrations of N_2O and NO_2 were negligible. In terms of NO emissions reduction, it was revealed that rich conditions were more effective. At the same time, pressure increases resulted in decreasing NO concentration in the post flame zone, as well as lower maximum concentration of NO and N_2O . Numerical analysis showed that N_2O and NO_2 were formed mainly from NO . To improve the agreement between experimental and numerical data, rate kinetic parameters of these reactions should be refined.

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1. Introduction

Fossil fuels are the basis of modern energy systems, sources that contribute to global warming, largely due to the formation of high CO_2 concentrations [1]. One solution to the problem of greenhouse emissions involves a transition to carbon-free fuels [2], and because its molecule has no carbon atoms, hydrogen is most promising. However, when made from renewable sources, its widespread use presents many challenges from high production costs. Another expense involves the necessity for complex and expensive equipment for its liquefaction and transportation. Hydrogen liquefies at the pressure of 350–700 atm, or at a temperature of -252.8°C , specific properties that make it unsuitable for wide use as a fuel.

Ammonia is considered to be an effective chemical hydrogen carrier [3], with a higher volumetric hydrogen density than liquid hydrogen [4]. Also, unlike hydrogen, ammonia becomes liquid at 8.5 atm pressure, at room temperature, or at atmospheric pressure and a temperature of -33°C . Such storage conditions are more ac-

ceptable for the potentially wider use of ammonia as a hydrogen carrier. Also, a well-established infrastructure for ammonia storage and transportation already exists.

As an individual fuel, ammonia has several problems however, including low burning velocity (around five times lower than methane), narrow flammability limits and high minimum ignition energy, and - due to the fuel nitrogen - a large amount of nitrogen oxide forms during ammonia combustion. Hydrogen is the most effective additive to improve the combustion characteristics of ammonia, and it can be produced via ammonia cracking. Many studies have focused on ammonia combustion and oxidation. This review mainly examines works on the effect of hydrogen addition on ammonia combustion.

As noted, one of the main problems of ammonia use is its low burning velocity. According to data reported in works by Takizawa et al., and Hayakawa et al. [5,6], the maximum flame speed of ammonia-air flames is 7 cm/s (atmospheric pressure, room temperature). The effect of hydrogen addition on the laminar flame speed of ammonia was examined in the work [7], and it was found that the addition of hydrogen significantly increased ammonia flame speed, while the greatest effect was observed in rich flames. At the same time, the dependence of the flame speed on hydrogen concentration was nonlinear: the burning velocity increased

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exponentially [8,9] with the increase of hydrogen fraction. A noticeable increase in flame speed was observed when the hydrogen fraction in the fuel blend was over 0.3 [10–12]. Nevertheless, the addition of hydrogen caused an increase in temperature in the post flame zone, which in turn, enhanced the formation of nitrogen oxides [7]. The pressure effect on the burning velocity of ammonia/hydrogen flames was studied by Wang et al. [13]; their work revealed that this effect was nonlinear. As expected, the higher temperature increased the flame speed of the ammonia/hydrogen flames [14]. Mei et al. [15] measured the laminar flame speed of $\text{NH}_3/\text{H}_2/\text{N}_2/\text{air}$ blends at pressures up to 10 atm. They revealed that laminar burning velocity increases with the increase of oxygen fraction and decreases with increasing pressure. In the work by Shrestha et al. [16] NH_3/air and $\text{NH}_3/\text{H}_2/\text{air}$ blends were used to compare the effect of oxygen and hydrogen concentration on laminar flame speed. The results show that the addition of 9% of O_2 (relatively to air) has the same effect as the addition of 30% of H_2 . Gotama et al. [17] also measured the laminar burning velocity of $\text{NH}_3/\text{H}_2/\text{air}$ flames and calculated the Markshtein lengths, which was shown to have a monotonous dependence on equivalence ratio. This dependence is mainly due to thermo-diffusive effects, which depends on the effective Lewis number and Zeldovich number of the flame.

Another important characteristic of fuels is ignition delay time, adding hydrogen to ammonia reduces this value. According to Shu et al. [18] a significant reduction in ignition delay time is achieved when the hydrogen content in an ammonia/hydrogen blend is at least 10%. However, the work of Dai et al. [19] showed that even 5% of hydrogen offered a tenfold reduction in ignition delay time. Mathieu and Petersen [20] showed that 5% of hydrogen decreased the autoignition temperature of ammonia/hydrogen blends by 100 K.

The effect of hydrogen addition on ammonia oxidation in a jet-stirred flow reactor was studied in the work of Sabia et al. [21], who found that increasing of ammonia concentration in the fuel blend shifts the temperature of intensive hydrogen oxidation towards higher temperatures. For example, the addition of 10% NH_3 shifted the onset temperature of $\text{H}_2/\text{O}_2/\text{Ar}$ and $\text{H}_2/\text{O}_2/\text{N}_2$ blends oxidation by about 50–75 K towards the higher temperature region. The effect of hydrogen addition on ammonia oxidation in a jet-stirred reactor at atmospheric pressure was examined in a recent paper by Zhang et al. [22]. Hydrogen content in the fuel blends was as high as 70%. It was revealed that hydrogen addition decreased the onset temperature of ammonia oxidation. The work of Osipova et al. [23] also reported the experimental data on ammonia/hydrogen blends oxidation in a jet-stirred reactor. Their experiments were also performed at atmospheric pressure and showed that the addition of 30% of hydrogen shifted the onset temperature of ammonia oxidation by ~ 250 K towards the lower temperature region. At the same time, the effect of equivalence ratio was negligible.

Only two works have presented experimental data on the flame structure of ammonia/hydrogen flames. The data on the flame structure of ammonia/hydrogen/oxygen/argon flames at low pressure (50–120 mbar) was presented by Duynslaegher et al. [24]. The authors showed that hydrogen presence in the fuel blend did not cause notable change in the composition of combustion products. However, in the hydrogen containing flames, the flame front shifted towards the burner surface. This work also showed that the transition to lean flames resulted in higher concentrations of NO in the post flame zone. A paper by Osipova et al. [25] presented experimental data on the structure of atmospheric pressure ammonia/hydrogen/oxygen/argon flames with different stoichiometry. According to their reported data, the main nitrogen containing species present in the post flame zone were molecular nitrogen and nitrogen oxide, while the concentration of N_2O was very low.

The transition from $\phi=0.8$ to $\phi=1.2$ made possible a decrease of NO concentration in the post flame zone, by more than one order of magnitude.

The potential application of ammonia as a fuel requires the development of comprehensive detailed chemical kinetic mechanisms, and many papers have presented such mechanisms. Miller and Bowman determined the main reactions involving ammonia and its radicals [26]. This mechanism was tested on experimental data obtained at low pressures, and it showed good predictive capability [27]. The mechanism proposed in the work of Lindstedt et al. [28] was also tested on the low-pressure (5 kPa) experiments. The main discrepancies between the experimental and numerical data were observed for NO concentrations in stoichiometric and rich flames. The main goal of the paper [29] was to develop the mechanism considering the effect of pressure on the fuel oxidation rate. Unlike previous works, high-pressure (30 and 100 atm) experimental data from the flow reactor was used to validate the mechanism; in general, modeling results were consistent with the experiments. Glarborg et al. [30] also developed the mechanism of ammonia oxidation. The rate constant for chain-branching reaction ($\text{NH}_2 + \text{NO} = \text{NNH} + \text{OH}$) was calculated using a high-level quantum chemistry approach. Experimental data on ammonia/hydrogen blends oxidation was used to examine the predictive capability of the mechanism from the work [31]. This data was obtained for free- and burner-stabilized flames, as well as for experiments in shock tubes, flow reactors and jet-stirred reactors. In general, the mechanism demonstrated good agreement with the experimental data. Rate constants of reactions involving NH_2 , HNO and N_2H_2 in the mechanism from work [29] were refined by Otomo et al. [32]. The resulting mechanism agreed well with the experimental data on laminar flame speed and ignition delay time obtained for a wide range of equivalence ratios and pressures. Validation of the mechanism proposed in [33] was performed using the experimental results for the flame structure at low pressure [24]. As noted, further refinement of the rate constants of reactions involving NH_2 and N_2O is still necessary. The chemical-kinetic mechanisms of ammonia oxidation were also presented in the works [34–36]; these models were validated specifically on the results of the structure of atmospheric pressure flames [37]. The mechanism of Mendiara and Glarborg [34] demonstrated the best predictive capability. The work [22] also proposed a detailed mechanism of ammonia combustion and oxidation. This mechanism was tested on a wide range of experimental data obtained for laminar flame speed, oxidation in flow and jet-stirred reactors, shock tubes, and burner-stabilized flames. In general, modeling results were consistent with the entire set of experiments. Okafor et al. [38] also presented a mechanism for ammonia combustion. This mechanism is based on GRI Mech 3.0 [39], and the reactions of ammonia oxidation were taken from the model developed by Tian et al. [40]. The resulting mechanism was tested on flame speed experiments and provided the satisfactory agreement. Another detailed mechanism of ammonia combustion was developed by Mei et al. [41]. The H_2 submechanism was taken from the work by Hashemi et al. [42], and NH_3 submechanism - from the work by Shrestha et al. [31]. The authors also updated the rate constants of such important reactions as $\text{H} + \text{O}_2 + \text{H} = \text{H}_2 + \text{O}_2$, $\text{H} + \text{O}_2 + \text{H} = \text{OH} + \text{OH}$, $\text{H} + \text{O}_2 + \text{O} = \text{OH} + \text{O}_2$, $\text{H} + \text{O}_2 + \text{OH} = \text{H}_2\text{O} + \text{O}_2$, $\text{NH}_2 + \text{H} = \text{NH} + \text{H}_2$, $\text{N}_2\text{H}_2 + \text{H} = \text{NNH} + \text{H}_2$, $\text{N}_2\text{H}_2 + \text{M} = \text{NNH} + \text{H} + \text{M}$. The developed model was validated on the experimental data obtained for laminar flame speed, ignition delay times as well as speciation in jet-stirred reactor and in low-pressure flames. This mechanism demonstrated good predictive capability for the entire set of experiments. A detailed chemical kinetic mechanism of ammonia oxidation was also proposed by Stagni et al. [43]. The H_2/O_2 submechanism was taken from the work of Metcalfe et al. [44],

the reaction of nitrogen-containing species - from the work of Song et al. [45]. The reactions of HONO/HNO₂ were updated based on the recommendation provided by Chen et al. [46]. Thermodynamic data were adopted from the database [47]. Experimental data on laminar flame speed and the speciation in jet-stirred and flow reactors were used for mechanism validation. An optimized version of Stagni mechanism [43] was presented in the work by Bertolino et al. [48]. The optimized version was tested on experiments on ammonia oxidation in the flow reactor and shock tube as well as against experiments on laminar flame speed and ignition delay times. In general, the optimized mechanism showed satisfactory agreement with the experimental results. The work by Li et al. [49] focused on the development of the reduced mechanism of ammonia combustion and oxidation. AramcoMech 2.0 [50], Shrestha model [31] and Tian model [40] were taken as the initial mechanisms. These models were combined and the reduction procedure was performed. The comparison with experimental data indicated that the reduced mechanism correctly reproduces the data on ignition delay times, however, overpredicts the values of laminar flame speed under rich conditions.

In the work by Mei et al. [15] a detailed mechanism of ammonia oxidation was also proposed. A previously developed mechanism [51] was taken as a base. The new model contains the updated rate constants of the following reactions: $\text{NH}_3 = \text{NH}_2 + \text{H}$, $\text{HNO} = \text{H} + \text{NO}$ [43], $\text{NH} + \text{NO} = \text{N}_2\text{O} + \text{H}$ [52], $\text{NH}_2 + \text{O} = \text{HNO} + \text{H}$ [53]. The resulting mechanism showed the good predictive capability for the laminar flame speed of ammonia flames. The work by Shrestha et al. [16] also focused on the development of detailed chemical-kinetic mechanism of ammonia combustion and oxidation. This mechanism is based on the model, which was suggested by the authors earlier [31]. In the new version, the rate constants of the reactions of NH_3 , NH_2 , NH , N_2H_2 , and H_2NO were updated. The experimental data on laminar flame speed were taken for the mechanism validation. In general, the satisfactory agreement between experimental and modeling results was achieved. Gotama et al. [17] also proposed the detailed mechanism of ammonia oxidation, which contains the updated rate constants of ammonia-radicals reactions. The proposed mechanism showed good agreement with the experimental data on the laminar burning velocity of $\text{NH}_3/\text{H}_2/\text{air}$ flames. Han et al. [54] also developed the model of ammonia combustion and oxidation. The base model was taken from the work [55], in which the submechanism of H/N/O was updated according to the recommendation from the works [56–58]. The proposed mechanism showed good predictive capability for the laminar flame speed and ignition delay times.

The above review showed that there are no current data on the structure of ammonia/hydrogen flames at high pressures, therefore, validation of the mechanisms on such experiments is not possible. Nevertheless, such experiments are extremely important for accurate description of chemical processes occurring at elevated pressures. The main purpose of this study is therefore to obtain new experimental data on the structure of ammonia/hydrogen flames at 4 and 6 atm, and to test eight different published mechanisms. This work presents an analysis of the effect of the equivalence ratio on the nitrogen oxides concentration in the post-flame zone.

2. Experimental method

To measure the chemical structure of premixed ammonia/hydrogen/oxygen/argon flames ($\phi=0.8, 1.0, 1.2$) at 4 and 6 atm, a molecular beam mass-spectrometric setup with soft electron impact ionization was used. Because a detailed description of the setup, flame stabilization system, and the data processing procedure, are available in the works [59,60,36,61], only the main details will be presented here.

Flames were stabilized at high pressure on a flat Botha-Spalding type burner [62]. The burner temperature was maintained at 368 K throughout the experiments. The burner was a perforated brass disc, 9.7 mm in diameter, and 1.5 mm thick; the diameter of the holes was 0.2 mm, and the distance between the holes' centers was 0.28 mm. Ammonia is known to react with copper alloys forming compounds that can affect combustion and cause burner corrosion, therefore, the burner surface was coated with a thin layer of nickel, preventing the interaction of ammonia with brass. A micro-screw mechanism was used to move the burner in the vertical direction. The distance between the burner surface and the probe tip was controlled with a cathetometer; the measurement accuracy was ± 0.01 mm.

To measure the flame structure at 4 and 6 atm, the burner was placed in a high-pressure chamber, pressurized with nitrogen, making it possible to perform experiments at the pressure up to 10 atm. To provide optical access, the camera was equipped with a special window fixed with a side flange. For chamber pressurizing, nitrogen was supplied through this flange to prevent water condensation on the window, ensuring its transparency during the experiments. Pressure in the chamber was controlled by a diaphragm pressure regulator on the outlet line; the accuracy of the pressure setting was 1%. To control the pressure, the chamber was equipped with a safety valve and a pressure gauge. The upper flange of the chamber was attached to the sampling probe flange. The scheme of the construction of the high-pressure chamber is provided in the works [63,64]. In high-pressure experiments, it is necessary to ensure an optimal ratio of the flows of unburnt gas and nitrogen, which was used to maintain the pressure the high-pressure chamber. Gas flow was set with mass flow controllers (Bronkhorst). The uncertainty of the gas flow rate values is ± 0.01 cm³/s.

The measurement of the flame structure at high pressures requires careful choice of flame stabilization conditions, which are: the flame should be visually stable; the temperature in the post-flame zone should not exceed 2000 K (in order to use S-type thermocouples); the flame zone must remain wide enough for measurements with a probe with certain spatial resolution. Since in our setup there is no ignition system in the high-pressure chamber, and, therefore, it is not possible to ignite the flame at high pressure, it is necessary to choose such conditions (composition, gas velocity on the burner surface) at which flame is stable both at atmospheric and at elevated pressures.

In the present experiments, linear velocity of the unburnt gas on the burner surface at different pressures was selected so that the flames were not only visually flat and stable (both at atmospheric and high pressures), but also the temperatures in the post-flame zone were as close as possible for flames of different stoichiometry. This ensured accurate analysis of the pressure and stoichiometry effects on the flame structure, since in this case the effect of temperature was negligible. Table 1 displays the composition of fresh gas blends and the conditions of stabilization for flames studied in this work.

High-pressure experiments also require the probe with the small orifice and thin walls near it (it is quite difficult when quartz is used). At the same time, such a probe must provide sufficient spatial resolution. In the present study, flames were sampled with a conical quartz probe (inner opening angle of 40° and the orifice diameter on the probe tip - 0.04 mm). The wall thickness near the probe tip was 0.08 mm [65,66]. These geometric parameters enable to reduce thermal and aerodynamic perturbations introduced by the probe into the flame. The sampling system enabled measurement of the concentration of stable species, as well as short-lived radicals. Gas sampled by the probe formed a molecular beam, which passed through the skimmer, modulator, and collimator, and entered the ion source of the quadrupole mass spectrometer. Soft electron impact ionization minimized the effect of ion fragmenta-

Table 1Molar composition of fresh gas blends and flame stabilization conditions in NH₃/H₂/O₂/Ar flames.

ϕ	p, atm	Reactants mole fraction				Gas velocity at burner surface (cm/s) T ₀ =368K	NH ₃ /H ₂ (molar ratio)
		NH ₃	H ₂	O ₂	Ar		
0.8 ± 0.02	4	0.077	0.077	0.121	0.725	7.72	1/1
1.0 ± 0.02	4	0.086	0.086	0.108	0.719	7.72	1/1
1.2 ± 0.02	4	0.106	0.106	0.11	0.679	7.72	1/1
0.8 ± 0.02	6	0.077	0.077	0.121	0.725	5.15	1/1
1.0 ± 0.02	6	0.086	0.086	0.108	0.719	5.15	1/1
1.2 ± 0.02	6	0.106	0.106	0.11	0.679	5.15	1/1

Table 2

Measured species. IE - ionization energy (reference data). E - energy of ionizing electrons.

m/z	Formula	Species name	IE (eV)	E (eV)	Calibration	Accuracy
1	H	atomic hydrogen	13.6	16.65	Direct	± 50%
2	H ₂	hydrogen	15.43	18	Direct	± 20%
16	NH ₂	amino radical	10.78	13.6	RICS (NH ₃)	± 50%
16	O	atomic oxygen	13.62	16.65	Direct	± 50%
17	NH ₃	ammonia	10.07	12.5	Direct	± 20%
17	OH	hydroxyl radical	13.02	16.65	Direct	± 50%
18	H ₂ O	water	12.62	16.65	O-balance	± 20%
28	N ₂	nitrogen	15.58	16.65	Direct	± 20%
30	NO	nitric oxide	9.26	18	Direct	± 20%
32	O ₂	oxygen	12.07	16.65	Direct	± 20%
40	Ar	argon	15.76	16.2	Direct	± 20%
44	N ₂ O	nitrous oxide	12.89	18	Direct	± 50%
46	NO ₂	nitrogen dioxide	9.59	18	RICS*(N ₂ O)	± 50%

*RICS - relative ionization cross-section.

tion for most of the measured peaks. The energy of ionizing electrons was selected individually for each species. The electron scattering energy was ±0.25 eV. Concentration profiles in this work were measured for the following species: H₂, NH₃, O₂, N₂, H₂O, NO, N₂O, NO₂, H, O, OH and NH₂. Table 2 shows the values of the energy of ionizing electrons, as well as the calibration method for each species.

Energy-dependent calibration coefficients were used to convert signal intensity in mass-spectrum (I_i) for the species i into its mole fraction (X_i). These coefficients were also normalized by the argon sensitivity coefficient (S/S_{Ar}). Thus:

$$I_i/I_{Ar} = S_i/S_{Ar} * X_i/X_{Ar}$$

Since the argon fraction in the mixtures studied was sufficiently large, it was considered to be nearly constant. Calibration coefficients for H₂, NH₃, O₂, N₂, NO and N₂O were obtained in direct calibration experiments, using gas blends of known composition. Calibration coefficients for H, OH and O were obtained using the same method performed in [67]. The calibration coefficient for H₂O was derived from the O-balance equation. Sensitivity coefficients for NO₂, and NH₂ were calculated using a method of relative ionization cross-section. Cross-section for NO₂ was taken from the NIST database [68]. Since the values of ionization cross-sections for NH₂ at different ionizing electron energies are not presented in this database, they were estimated using methods described in the papers [69,70]. Measurement accuracy of the concentration of reagents and the main combustion products was ± 20% of their maximum concentration in the flame. For the other species, experimental uncertainty was ±50%.

It is known, that high-pressure experiments require thin thermocouples, which enable to measure the temperature profiles in flames with the narrow zone of the main chemical reactions. In this work, temperature profiles were measured with thin S-type thermocouples (Pt/Pt + 10%Rh), made from wire of 30 μm in diameter. It is important that for the measurement of the temperature profiles in the probe presence, the thermocouple junction must be

position under the probe orifice at the certain space point. In our experiments, the thermocouple junction was located at a distance of 0.2 mm from the probe tip. To avoid catalytic reactions on the thermocouple surface, it was coated with a thin layer of SiO₂. A detailed description of the thermocouple design and temperature measurement procedures can be found in [71]. To consider the probe effect on the flame (i.e., the effect of gas dynamic perturbations), experimental profiles of the species were shifted towards the burner surface [60]. For each flame, the shift value was the same for all species. In this work, the shift value was ~0.2 mm.

The main sources of the experimental uncertainties at elevated pressures and their limits are the following:

- 1) The accuracy of the unburnt gas composition (the system for the preparation of fresh gas blend enables to set the volumetric flow rates with the accuracy of ±0.01 cm³/s at T = 298 K and p = 1 atm).
- 2) The accuracy of the pressure value inside the chamber (in our experiments ±0.05 atm)
- 3) The accuracy of the determining the initial distance between the burner and the probe (±0.01 mm)
- 4) Condensation of combustion products in the chamber (water vapor), which can subsequently complicate flame sampling and cause the instability of the measured species concentration in time

3. Modeling

The PREMIX code from CHEMKIN package was used for numerical modeling of the flames structure. The values of GRAD and CURV were equal to 0.05 and 0.05, respectively; multi-component and thermal diffusion options were considered. Experimentally measured temperature profiles were used as input data for simulation, making it possible to consider probe effects. Eight chemical kinetic mechanisms were tested in the present work. Table 3 lists the basic information regarding the applied mechanisms.

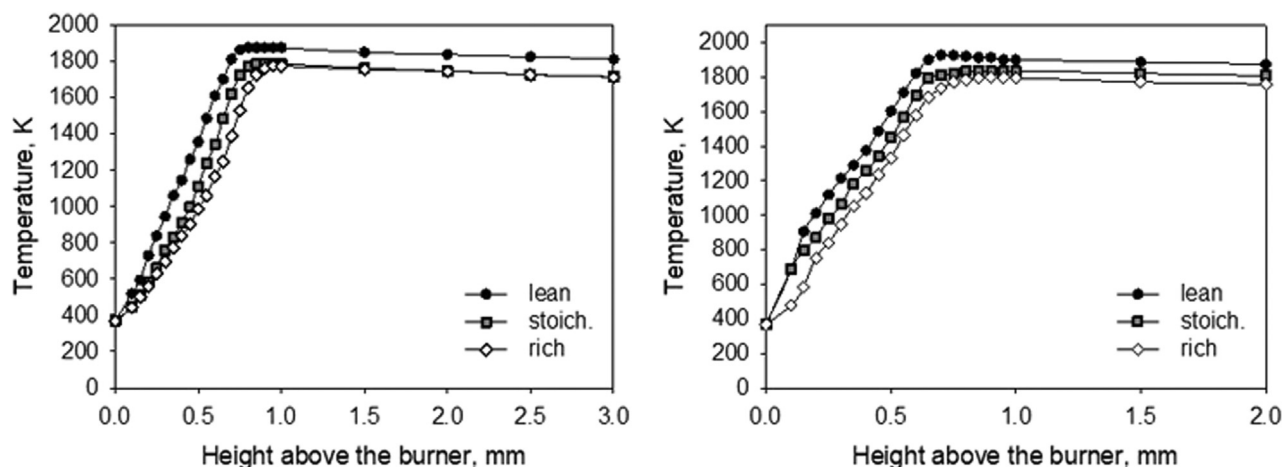


Fig. 1. Temperature profiles for lean, stoichiometric, and rich $\text{NH}_3/\text{H}_2/\text{O}_2/\text{Ar}$ flames at 4 (left) and 6 atm (right).

Table 3

Kinetic mechanisms used for the $\text{NH}_3/\text{H}_2/\text{O}_2/\text{Ar}$ flame modeling.

Model №	Species number	Reactions number	Year	Reference
Model 1	151	1397	2018	[30]
Model 2	59	356	2018	[38]
Model 3	127	1207	2010	[33,36]
Model 4	38	263	2021	[22]
Model 5	40	257	2021	[15]
Model 6	31	203	2021	[48]
Model 7	125	1099	2021	[16]
Model 8	36	298	2021	[54]

4. Results

4.1. Structure of $\text{NH}_3/\text{H}_2/\text{O}_2/\text{Ar}$ flames at 4 and 6 atm

This section presents experimental and numerical data for lean ($\phi=0.8$), stoichiometric ($\phi=1.0$) and rich ($\phi=1.2$) ammonia/hydrogen flames at 4 and 6 atm. Figure 1 shows experimentally measured temperature profiles.

At 4 atm, in the lean flame, it can be seen that the reaction zone was approximately 0.6–0.7 mm. In a stoichiometric flame, the zone of the main chemical reactions increased slightly and was equal to 0.7–0.8 mm. The rich flame zone was 0.9 mm; the temperature in the post flame zone in the lean flame was approximately 1800–1900 K. At the same time, this temperature was ~1700–1800 K for stoichiometric and rich flames. These values indicated that the temperature in the post flame zone for the flame studied differed by no more than 100 K. At 6 atm, the reaction zone of the lean flame was 0.5–0.6 mm. For the stoichiometric flame, there was a slight increase of the flame zone width, to 0.6–0.7 mm. The reaction zone of the rich flame was 0.7–0.8 mm. The post flame zone temperature of the lean flame was ~1850–1950 K; for stoichiometric flame, this temperature was ~1800 K. For the rich flame, the temperature in the zone of combustion products was ~1750–1800 K. Similar to 4 atm experiments, the temperature in the post flame zone differed by no more than 100 K. The value of this difference was insignificant and made it possible to compare the data on the structure of $\text{NH}_3/\text{H}_2/\text{O}_2/\text{Ar}$ flames, obtained under nearly the same conditions.

Figure 2 presents the measured mole fraction profiles of the fuel blend components (NH_3 and H_2), as well as the modeling results obtained with the eight published mechanisms. These graphs show the thermodynamically equilibrium mole fractions.

A comparison of experimental and numerical data showed that, in general, all the mechanisms agreed well with the measurement results. The experimental and numerical data in Fig. 2 indicate that in lean flame both ammonia and hydrogen were completely consumed in the post flame zone. In stoichiometric conditions, ammonia was also consumed completely, as seen from the experimental and numerical data (considering experimental uncertainty). At the same time, hydrogen was present in the post flame zone. According to the experimental data, hydrogen concentration in the post flame zone was slightly higher than all the models predicted. However, this discrepancy was not very significant. The calculated concentrations in the post flame zone were close to those obtained for conditions of thermodynamic equilibrium. In rich flame, a small amount of ammonia was present in the post flame zone, which was observed for both experimental and numerical data. The hydrogen profile had a slight inflection. Moreover, the transition from $\phi=1.0$ to $\phi=1.2$ resulted in an increase of hydrogen concentration in combustion products by about 2–4 times. Figure 3 shows experimental and numerical mole fraction profiles of O_2 and the main combustion products (H_2O and N_2) in $\text{NH}_3/\text{H}_2/\text{O}_2/\text{Ar}$ flames at 4 atm.

According to Fig. 3, experimental and numerical results of O_2 , N_2 and H_2O agreed within the experimental uncertainty. In the lean flame, O_2 remained in the post flame zone, whereas in stoichiometric and rich flames, oxygen was almost completely consumed, as expected. As for N_2 , the transition from lean to rich flame resulted in a slight increase of the concentration of molecular nitrogen in the post flame zone. N_2 concentration is $\sim 4 \times 10^{-2}$ in lean flame, $\sim 4.5 \times 10^{-2}$ in stoichiometric, and $\sim 5 \times 10^{-2}$ in rich flame. When ϕ increased, the concentration of available oxygen decreased; on the contrary, the concentration of available ammonia increased. This may serve as a possible explanation since ammonia is the main source of N_2 . For water concentration profile, the change of the equivalence ratio had only a modest effect on the water mole fraction in the post flame, which was $\sim 2 \times 10^{-1}$. The calculated concentrations in the post flame zone were close to thermodynamically equilibrium values. Figure 4 presents experimental and numerical data for mole fraction profiles of NO , N_2O and NO_2 in $\text{NH}_3/\text{H}_2/\text{O}_2/\text{Ar}$ flames at 4 atm.

Osipova et al. [25] showed that among NO_x , only NO was present in a significant concentration in the post flame zone. A similar trend was observed when the pressure increased up to 4 atm. There was good agreement between experimental and modeling results in the lean flame for almost all tested models, however Model 8 overpredicts NO concentration in the post flame zone and Model 6 and 7, on the contrary, underpredict this value. In the sto-

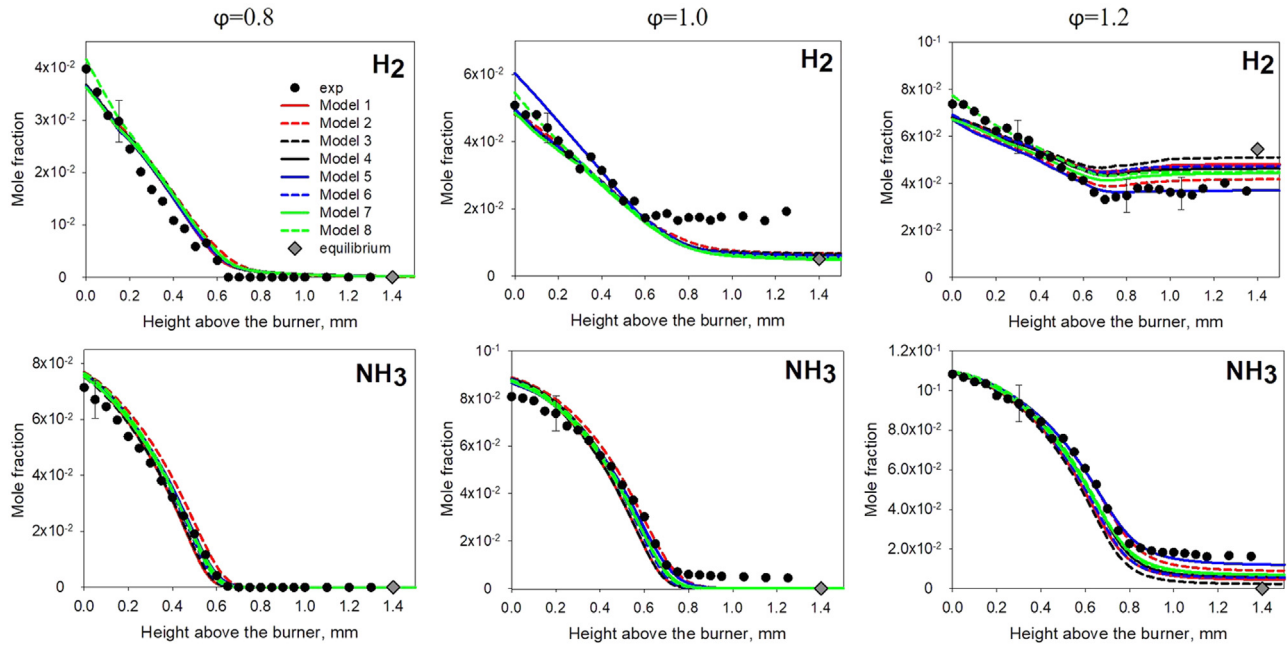


Fig. 2. Mole fraction profiles of H_2 and NH_3 for lean, stoichiometric, and rich $NH_3/H_2/O_2/Ar$ flames at 4 atm.

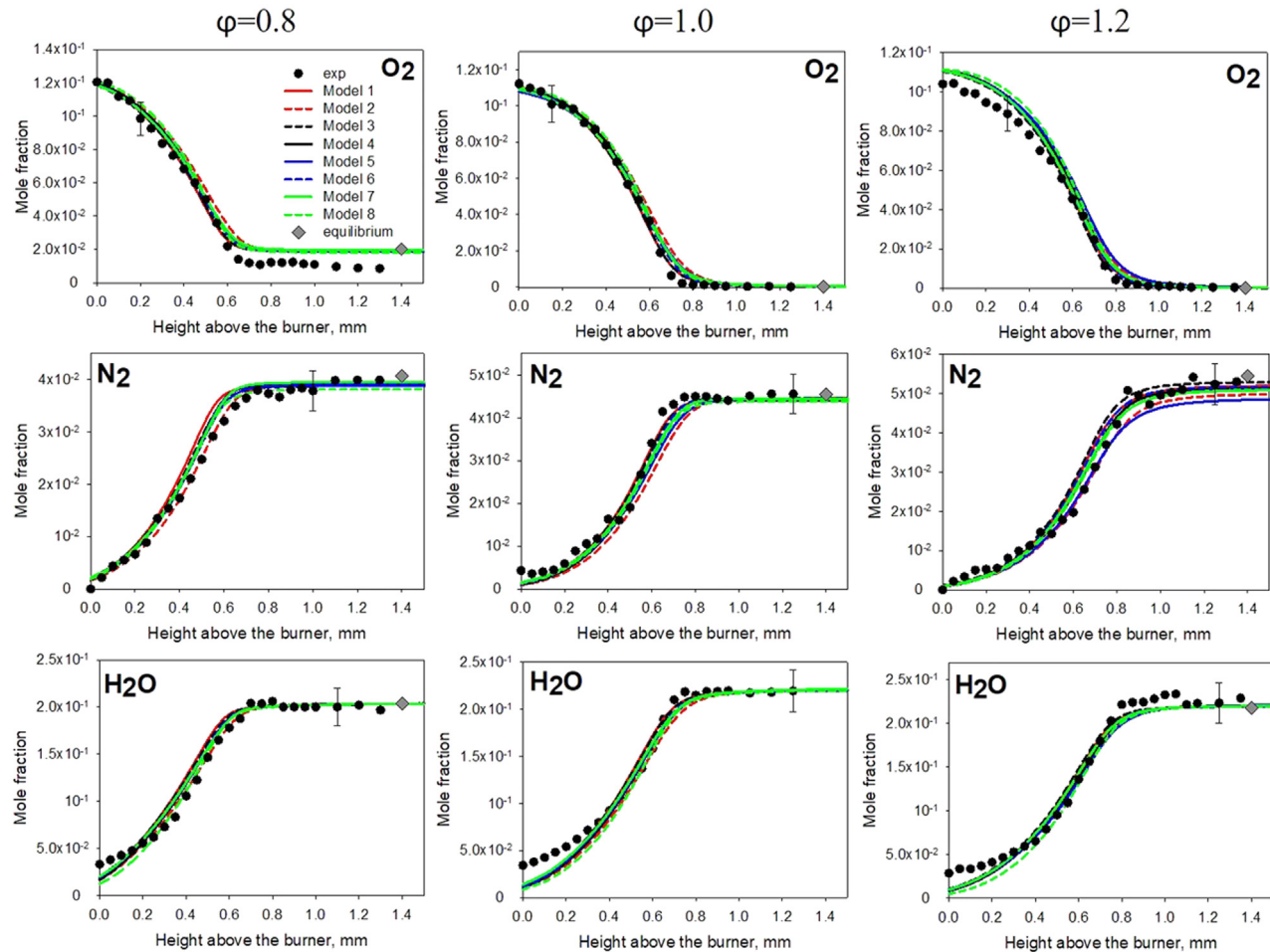


Fig. 3. Mole fraction profiles of O_2 , N_2 and H_2O for lean, stoichiometric, and rich $NH_3/H_2/O_2/Ar$ flames at 4 atm.

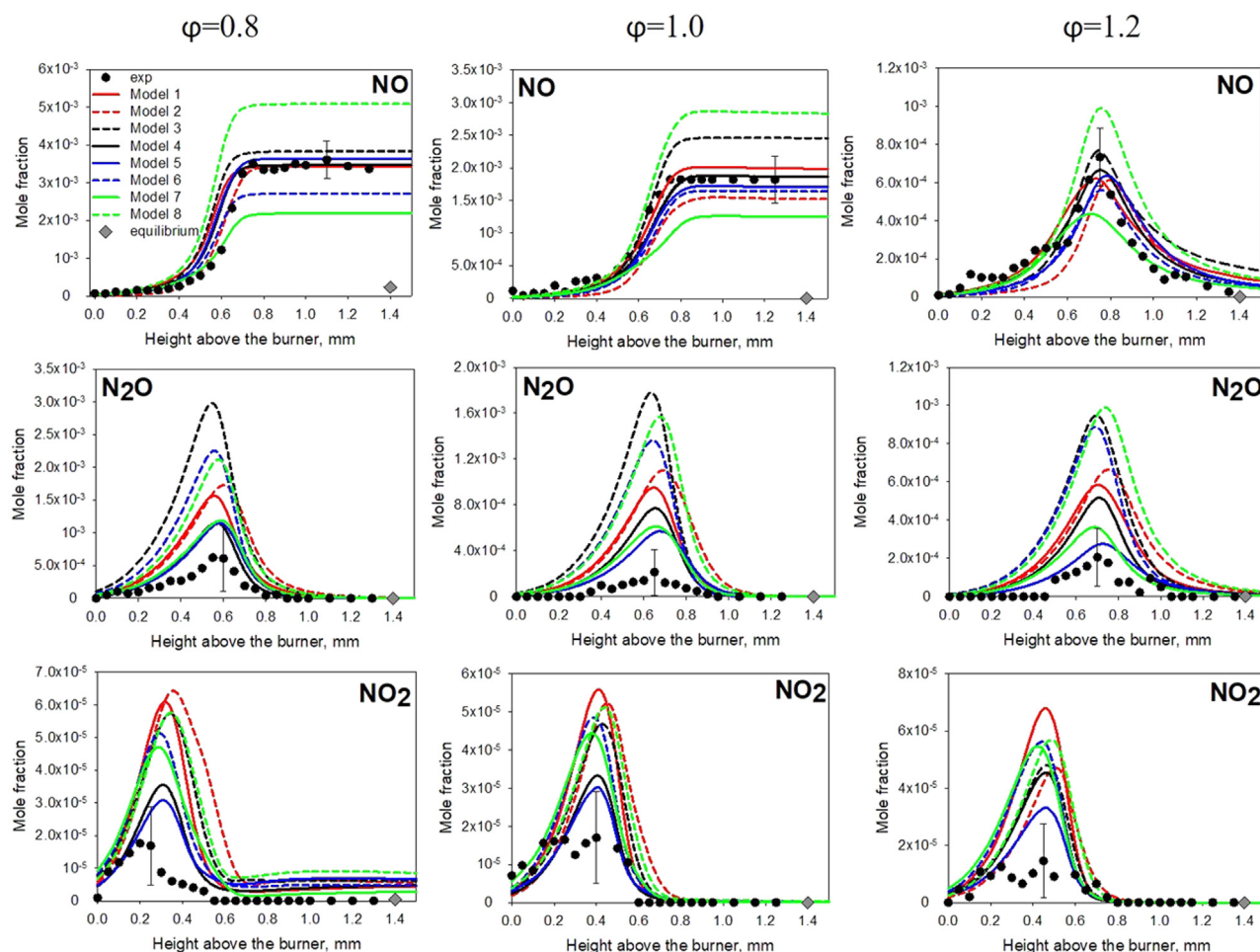


Fig. 4. Mole fraction profiles of NO, N₂O and NO₂ for lean, stoichiometric, and rich NH₃/H₂/O₂/Ar flames at 4 atm.

ichiometric flame, Model 3 and 8 overpredicted the concentration of NO in the post flame zone compared to the experiment, whereas Model 7 predicted the lower value. For the rich flames, there was good qualitative and quantitative agreement between experimental and numerical data, since the models correctly described the profile's shape at the peak position. All the models predicted nearly the same value of NO peak concentration, however, Model 8 overestimated the maximum fraction of NO. In the rich flame, the NO concentration on the post flame zone was close to its thermodynamically equilibrium value, while in the lean and stoichiometric flames, NO concentration in the post flame zone was significantly higher than the values in conditions of thermodynamic equilibrium.

In terms of agreement between experimental and numerical data, the trend for N₂O was nearly the same in all three flames. All the mechanisms adequately predicted the profile shape and position of the maximum concentration. However, for the absolute value of N₂O peak concentration there is more noticeable scatter than for NO. In the post flame zone, N₂O concentration was insignificant and equal to $\sim 10^{-7}$. The same trend was observed in the work of Osipova et al. [25]. N₂O concentration in the post flame was close to the thermodynamically equilibrium value.

In the case of NO₂, all the mechanisms adequately described both the shape of the concentration profile and the maximum position. Similar to N₂O, for the absolute value of the maximum concentration of NO₂ there is a larger scatter compared to NO. The experimental and numerical values of NO₂ mole fraction in the post flame zone were close to the calculated thermodynamically

equilibrium concentrations. Figure 5 shows the measured and simulated concentration profiles of H, O, OH and NH₂ radicals.

In general, all models correctly describe the profiles shape and the position of the maximum concentration of radicals. For the modeling values of H, O, OH and NH₂ peak concentration there is some scatter. The highest concentration of H-radical was observed in stoichiometric flame; in rich flame it was slightly lower, and the lowest concentration of H was in the lean flame. As expected, in the lean flame O-radical concentration had the highest value, and in rich flame conditions, the value was the lowest. The highest concentration of OH radical corresponded to lean conditions, since in this case more oxygen was available. The increase of equivalence ratio resulted in a slight increase of the maximum NH₂ concentration. All the models qualitatively reproduced this trend.

Figure 6 shows mole fraction profiles of the fuel blend components (NH₃ and H₂) obtained experimentally and numerically using eight detailed chemical-kinetic mechanisms at 6 atm. These graphs also show H₂ and NH₃ concentration in conditions of thermodynamic equilibrium.

Comparison of experimental and numerical data indicated that similarly to results at 4 atm, all the mechanisms agreed with the measured profiles. Like the results at 4 atm, in lean conditions, both ammonia and hydrogen are fully consumed. The increase of equivalence ratio results into the increase of hydrogen concentration in the post-flame zone. It should be noted that there is some scatter of the modeling values of H₂ mole fraction in the post flame zone in rich conditions. Figure 7 shows the experimental and numerical mole fraction profiles of O₂ and the main oxida-

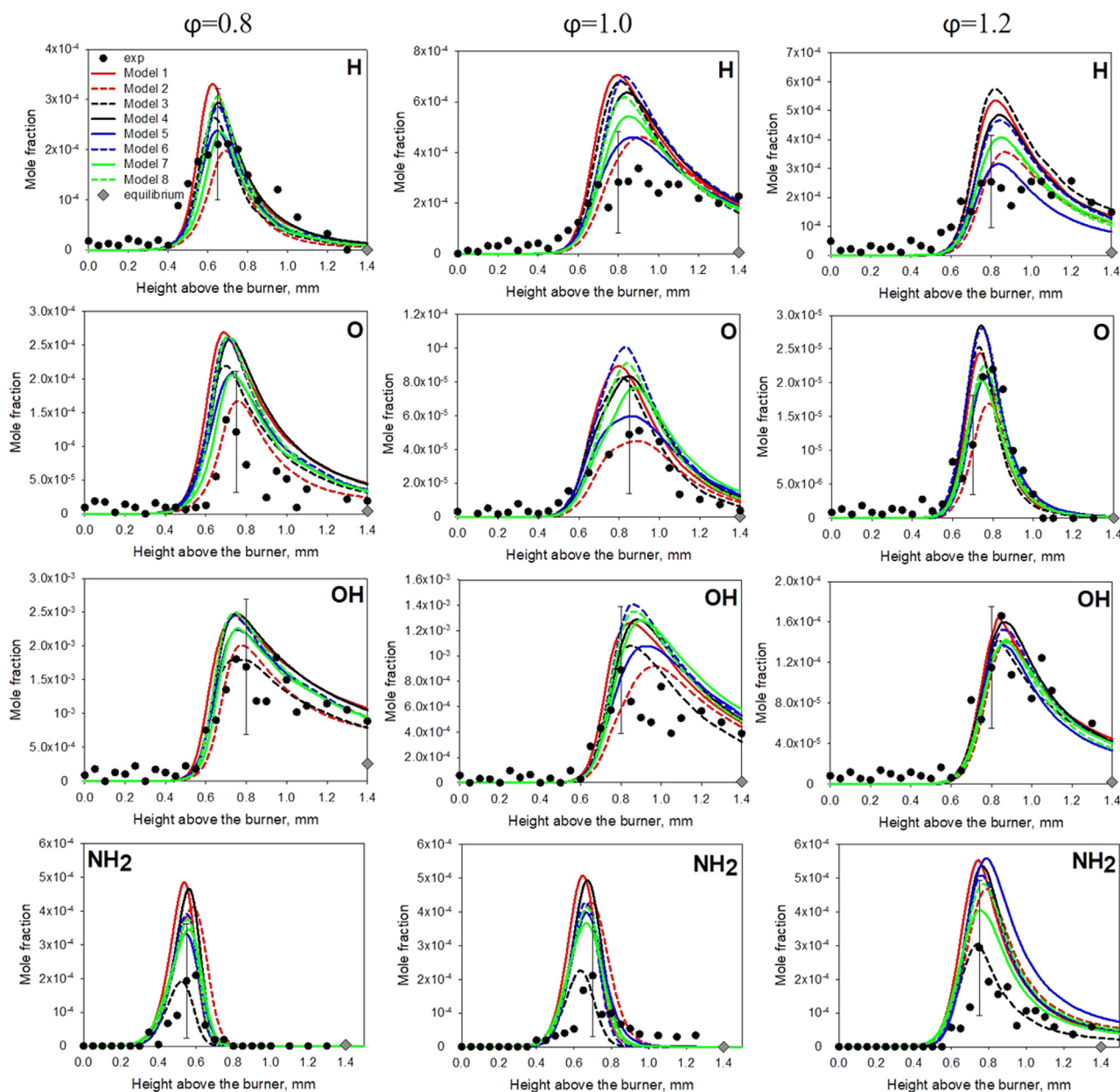


Fig. 5. Mole fraction profiles of H, O, OH and NH_2 -radicals in lean, stoichiometric and rich $\text{NH}_3/\text{H}_2/\text{O}_2/\text{Ar}$ flames at 4 atm.

tion products (such as H_2O and N_2) in the $\text{NH}_3/\text{H}_2/\text{O}_2/\text{Ar}$ flames at 6 atm.

For O_2 , H_2O and N_2 the experimental and numerical data agreed within the experimental uncertainty. The increase of equivalence ratio results in the decrease of O_2 concentration in the post-flame zone and in a slight increase of N_2 mole fraction. For H_2O there is very weak effect. The concentrations of N_2 and H_2O in the post flame zone were close to those under thermodynamically equilibrium conditions. Figure 8 shows experimental and numerical mole fraction profiles of NOx (such as NO, N_2O and NO_2) in $\text{NH}_3/\text{H}_2/\text{O}_2/\text{Ar}$ flames at 6 atm.

In general, for NOx there is the same trend of agreement of experimental and modeling data as in the case of 4 atm. Namely, NO concentration profile is correctly reproduced by almost all models, except Models 6, 7 and 8 in lean flame and Models 3, 7 and 8 in stoichiometric flame. Model 8 noticeably overestimates the peak NO concentration in the rich flame. For N_2O and NO_2 all models are able to reproduce the profiles shape and peak position. How-

ever, in terms of quantitative agreement there are larger discrepancies than for NO and some scatter in the values of peak concentrations. Figure 9 presents the measured and simulated concentration profiles of H, O, OH and NH_2 radicals at 6 atm.

In the case of reactive species at 6 atm there is almost the same trend as at 4 atm. All tested models capture the profiles shape and the peak position, however, there are some quantitative discrepancies between experimental and numerical data and the modeling values of the radicals' maximum values have scatter.

4.2. Analysis of nitrogen oxide formation in $\text{NH}_3/\text{H}_2/\text{O}_2/\text{Ar}$ flames

One of the most important issues of potential ammonia application is chemically bonded nitrogen in its molecule. Fuel-nitrogen inevitably causes increased NOx emissions, making it necessary to choose the optimal combustion mode to reduce NOx concentration for practical ammonia implementation. This section offers analysis of the equivalence ratio effect, as well as the effect of pressure

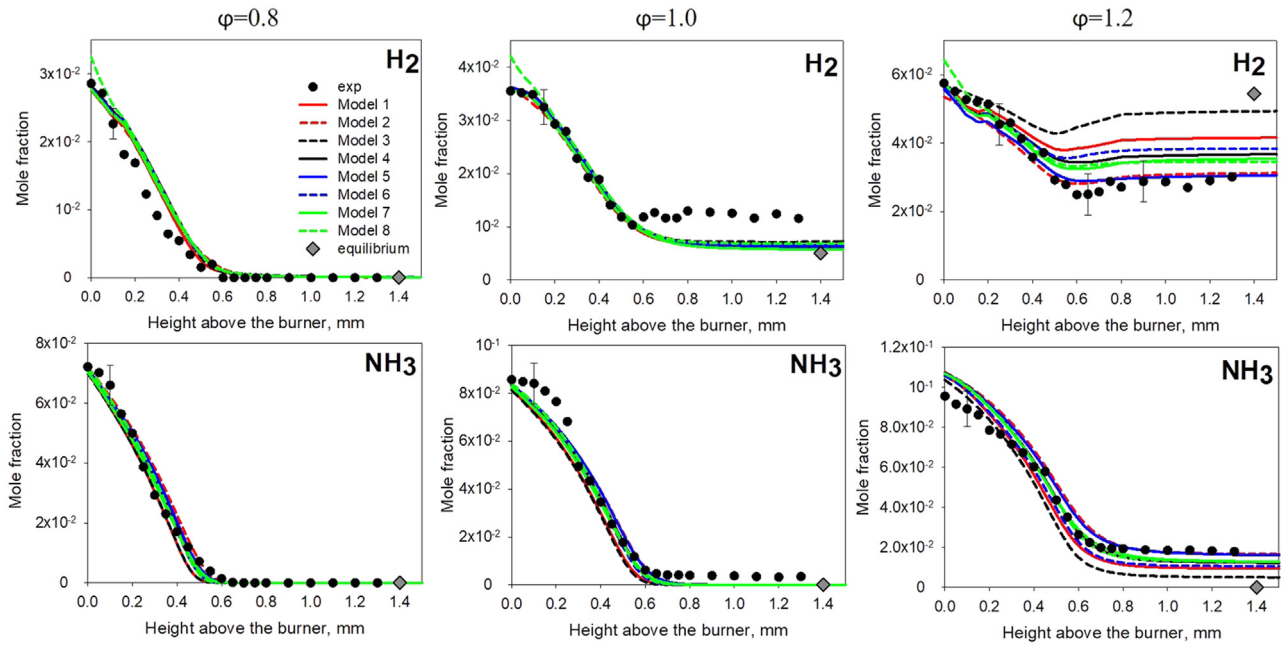


Fig. 6. Mole fraction profiles of H_2 and NH_3 for lean, stoichiometric, and rich $NH_3/H_2/O_2/Ar$ flames at 6 atm.

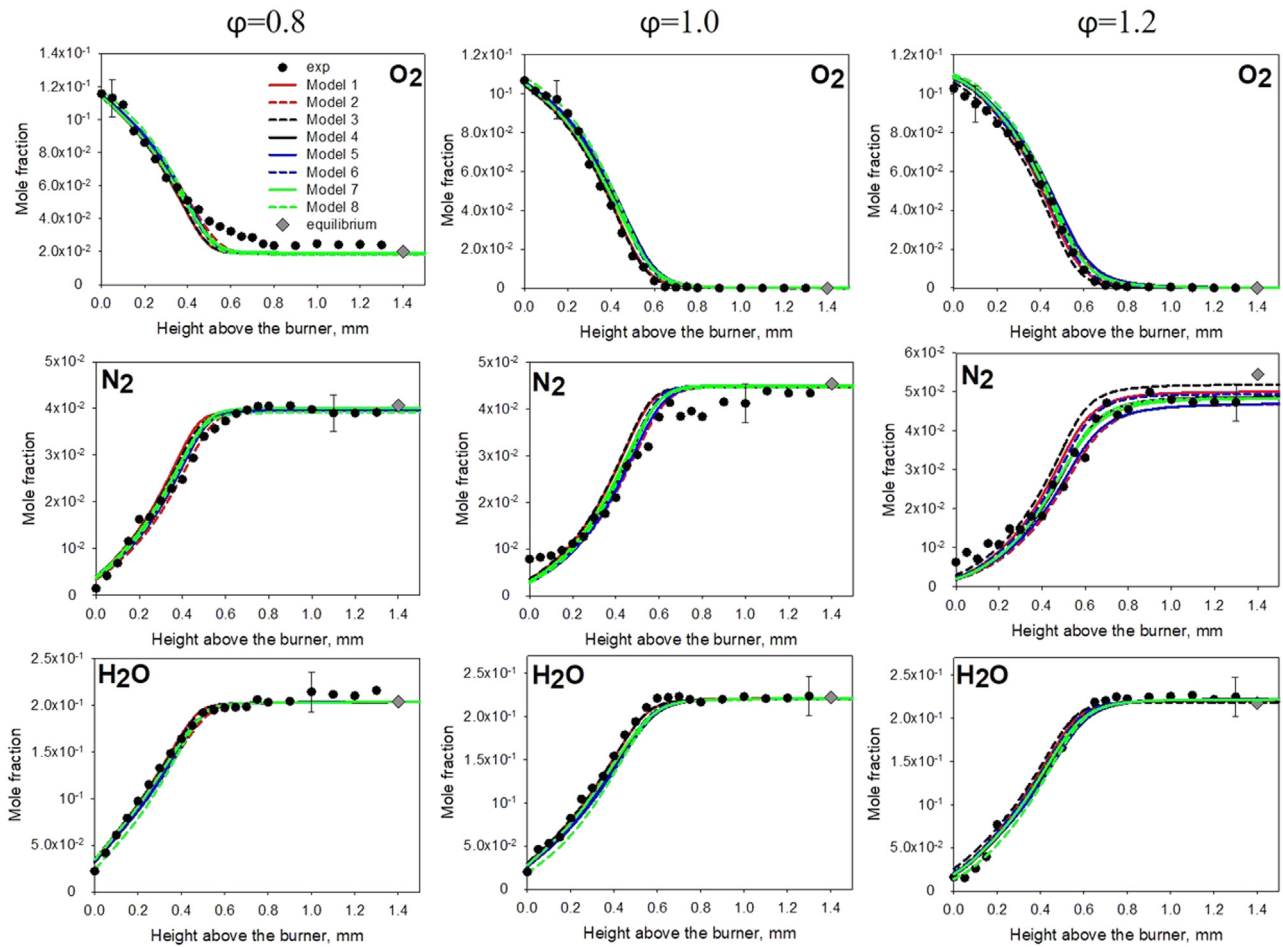


Fig. 7. Mole fraction profiles of O_2 , N_2 and H_2O for lean, stoichiometric, and rich $NH_3/H_2/O_2/Ar$ flames at 6 atm.

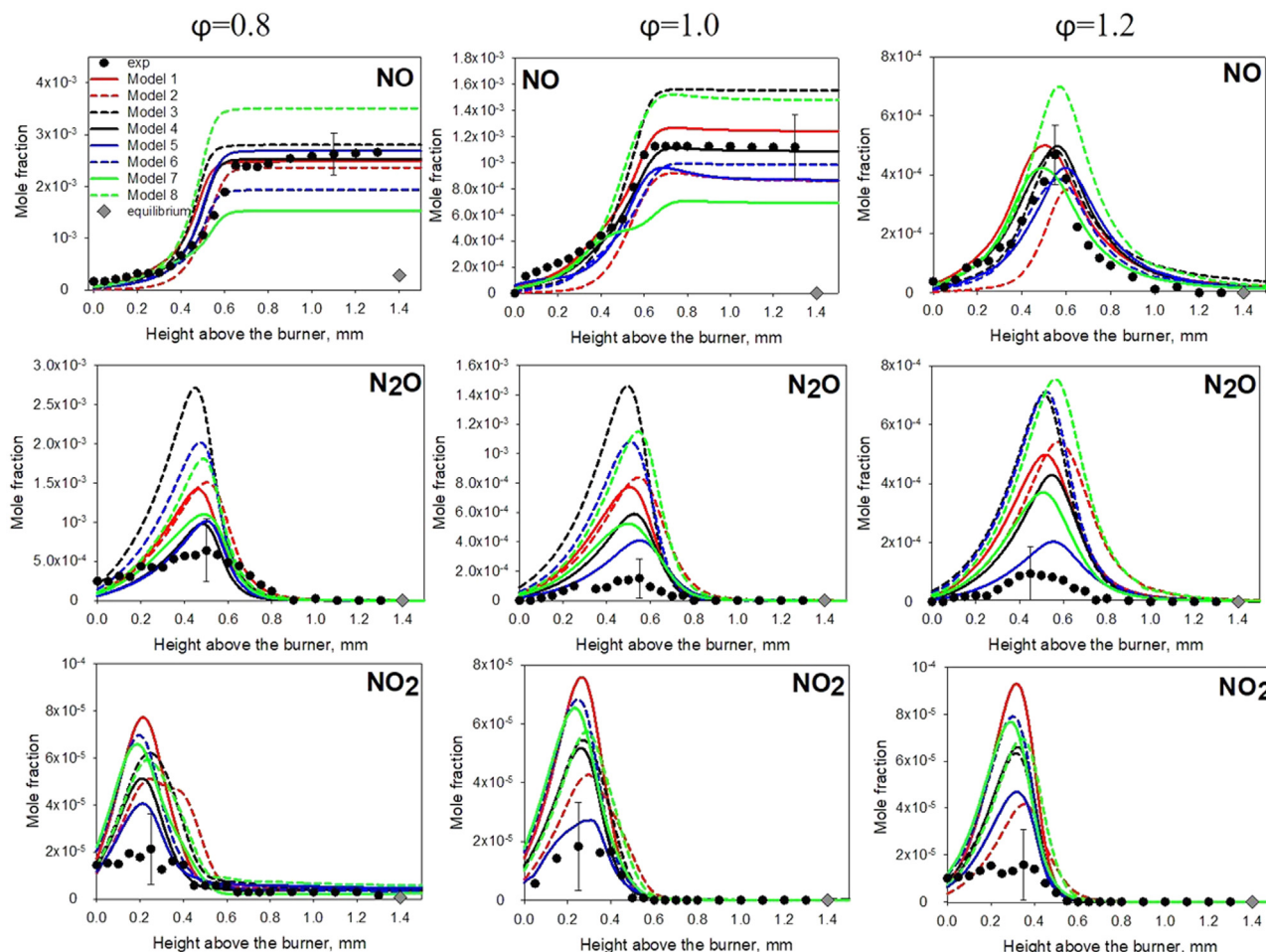


Fig. 8. Mole fraction profiles of NO, N₂O and NO₂ for lean, stoichiometric, and rich NH₃/H₂/O₂/Ar flames at 6 atm.

on NO_x concentration in the combustion products, and their maximum concentration in the zone of intensive NH₃ oxidation.

4.2.1. Effect of equivalence ratio at 4 and 6 atm

Figure 10 shows experimental and numerical values of NO concentration in the post flame zone, as well as the peak concentration of NO, N₂O and NO₂ in NH₃/H₂/O₂/Ar flames vs the equivalence ratio at 4 atm. The transition from $\phi=0.8$ to $\phi=1.0$ reduced the concentration of NO by nearly 1.8 times. In rich flame, the concentration of NO in the post flame zone was $\approx 10^{-6}$. All models reproduced this trend, although Model 3 slightly overestimated NO concentration in the post flame zone under stoichiometric conditions. The largest discrepancies are observed for Models 6, 7 and 8. This data indicates that rich blends were more effective in terms of NO concentration reduction (Fig. 10a).

However, Fig. 4 shows that in rich flame the mole fraction profile of NO had a peak. Since in real engines and burner devices, combustion products (corresponding to the zone of the main chemical reactions) can be quenched after contact with the relatively cold walls of the combustion chamber, it was necessary to pay particular attention to the effect of combustion conditions on the maximum concentrations of key species in the flames. Therefore, the effect of equivalence ratio on NO peak concentration should also be analyzed. Figure 10b shows experimental and numerical values of NO peak concentration in NH₃/H₂/O₂/Ar flame vs the equivalence ratio. In lean and stoichiometric flames, the values of the peak concentration, and the concentration of NO in the post

flame zone, were the same. The transition from $\phi=0.8$ to $\phi=1.0$ reduced NO concentration by about 1.8 times. The change of equivalence ratio from 1.0 to 1.2 decreased the peak concentration of NO by about 3.5 times. This trend was qualitatively reproduced by all the mechanisms, however, as in previous case, the largest discrepancies show Models 6, 7 and 8. Figure 10b shows that, even for the reduction of NO peak concentration, rich blends were more favored, since in rich conditions NO peak concentration was lower than in lean and stoichiometric.

As noted, the mole fraction profiles of N₂O and NO₂ were also measured in this work. Since the concentration of these species in the post flame zone was insignificant, this section provides analysis of the equivalence ratio effect on the peak concentration of N₂O and NO₂.

Figure 10c presents experimental and numerical values of N₂O peak concentrations in NH₃/H₂/O₂/Ar flames vs the equivalence ratio. It is shown that in the stoichiometric flame, the concentration of N₂O was around three times lower than in the lean flame. Further increase of the equivalence ratio to $\phi=1.2$ had only a very weak effect on the N₂O peak concentration. The best performance show Models 4, 5 and 7, whereas other curves have a greater slope to the horizontal axis.

Figure 10d presents experimental and numerical values of NO₂ peak concentrations in NH₃/H₂/O₂/Ar flames vs equivalence ratio. According to Models 1, 4, 5, 6 and 7 the transition from $\phi=0.8$ to $\phi=1.0$ did not change the peak concentration of NO₂. For $\phi=1.2$ there was a slight increase of NO₂ peak concentration. Models 2

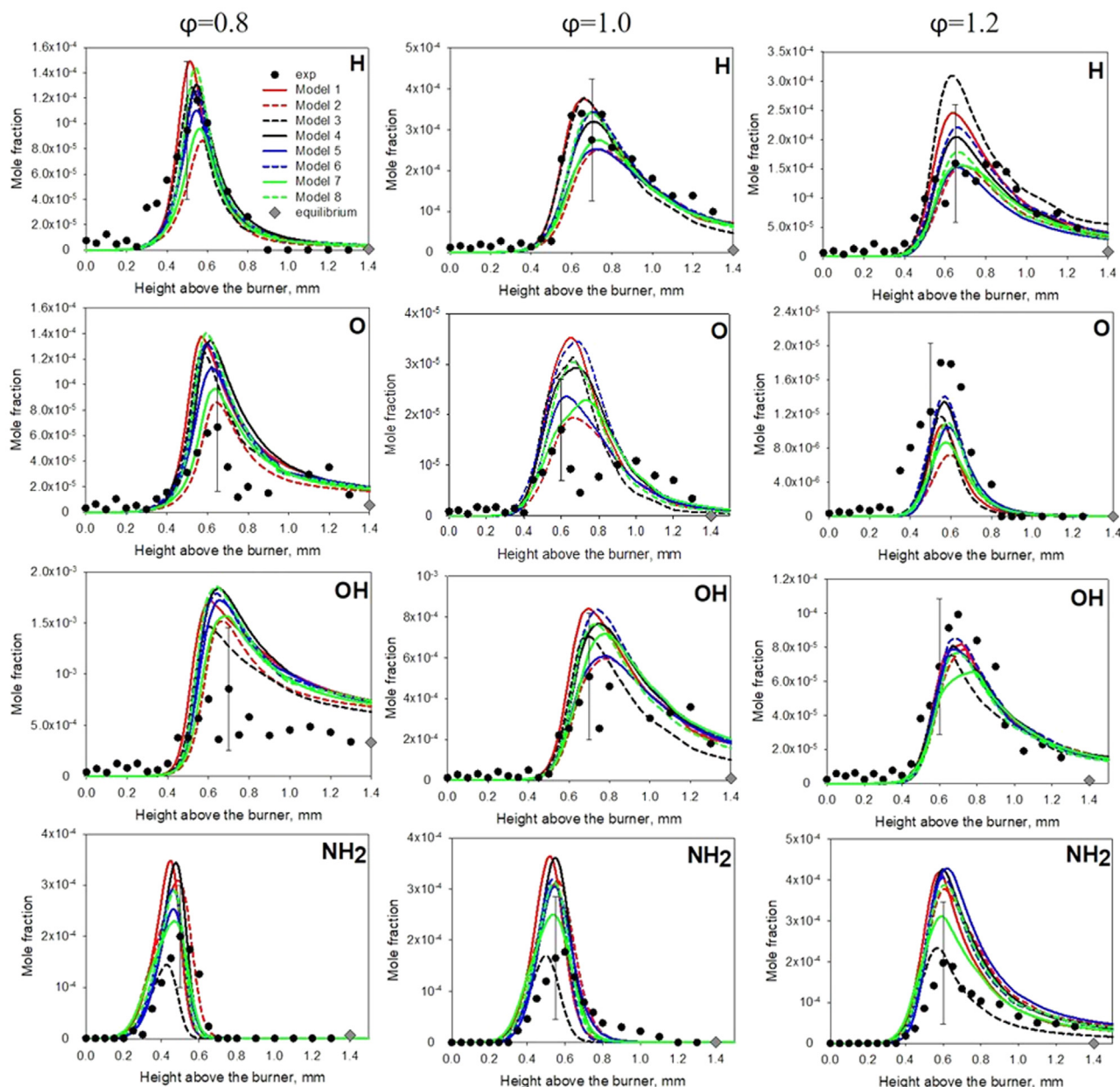


Fig. 9. Mole fraction profiles of H, O, OH and NH_2 -radicals in lean, stoichiometric and rich $\text{NH}_3/\text{H}_2/\text{O}_2/\text{Ar}$ flames at 6 atm.

and 3 predicted the opposite tendency; for these models, a decrease of NO_2 concentration was observed in the transition from lean to stoichiometric conditions, further increase of the equivalence ratio results into a very slight reduction of NO_2 concentration. According to Model 8 there is a slight decrease of NO_2 concentration when equivalence ratio changes from 0.8 to 1.0, and slight increase when equivalence ratio is 1.2. For this species, it was difficult to reach a conclusion about the quantitative agreement of experimental and numerical data, or about reproducing the dependence of NO_2 peak concentration on the equivalence ratio. In the experiment, the value of NO_2 peak concentration was $\sim 10^{-5}$, close to the sensitivity limit of the setup in this work.

Figure 11 shows experimental and numerical values of NO concentration in the post flame zone, as well as peak concentration of NO, N_2O and NO_2 in $\text{NH}_3/\text{H}_2/\text{O}_2/\text{Ar}$ flames vs equivalence ratio at 6 atm. For NO concentration, there is the same trend as in the case of 4 atm, namely, wherein the concentration of NO (both peak and in the post-flame) zone decreases with the increase

of equivalence ratio. All Models in general capture this dependence, however, Models 6, 7 and 8 show quantitative discrepancies (Fig. 11a, b).

Figure 11c shows the experimental and numerical values of the N_2O peak concentration in $\text{NH}_3/\text{H}_2/\text{O}_2/\text{Ar}$ flame vs equivalence ratio. As for results at 4 atm, the transition from lean to stoichiometric flame decreased the N_2O peak concentration by about three times. Further transition to rich conditions had a modest effect on the N_2O peak concentration. All the models qualitatively reproduced this tendency. Models 4, 5 and 7 show the best performance. Figure 11d depicts the experimental and numerical values of the NO_2 peak concentration in $\text{NH}_3/\text{H}_2/\text{O}_2/\text{Ar}$ flame vs equivalence ratio. As in the case of 4 atm, Models predict different behavior on NO_2 concentration vs equivalence ratio.

4.2.2. Effect of pressure

To analyze the effect of pressure, the data obtained at 1 atm with the same mass spectrometric setup were used [25]. Figure 12

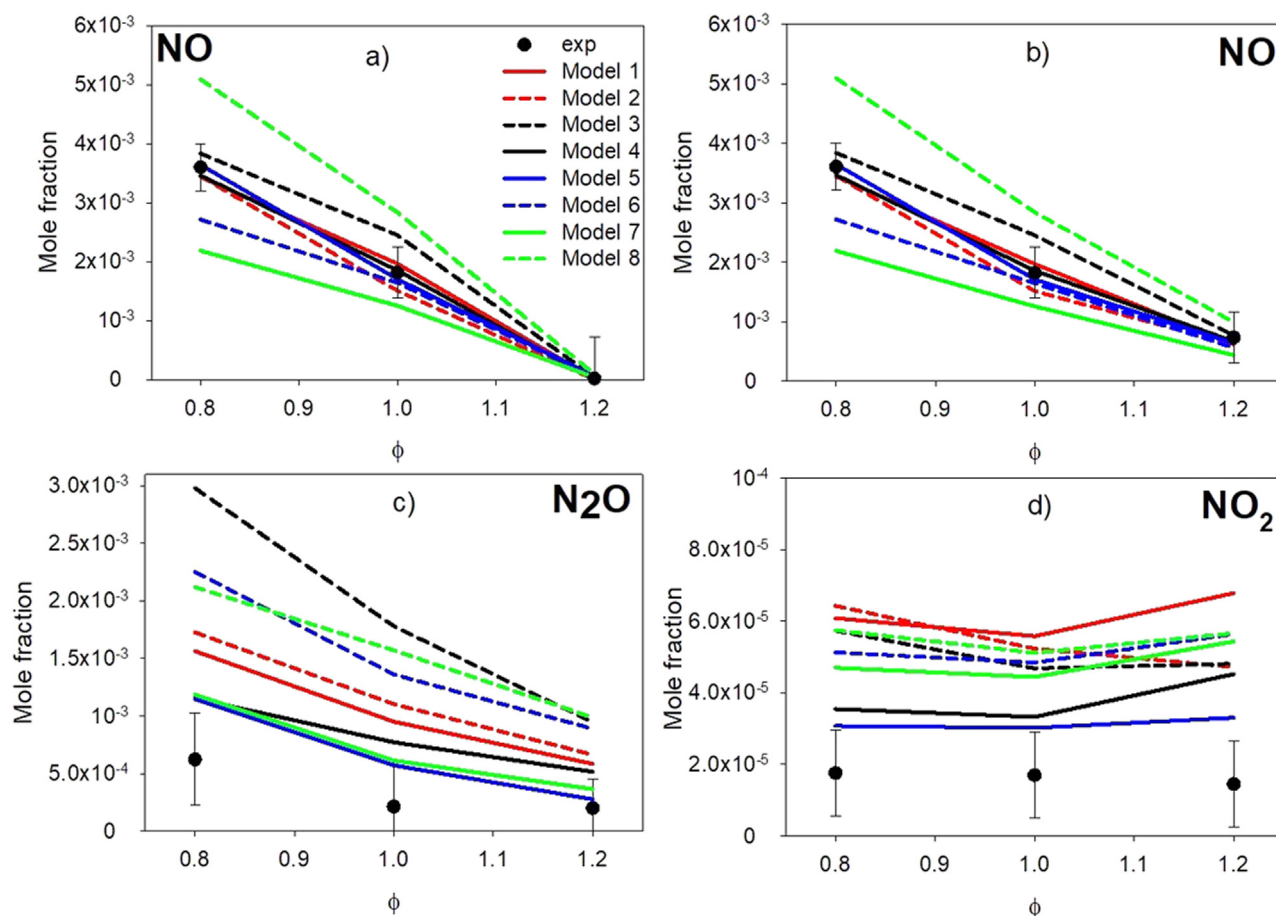


Fig. 10. Experimental and numerical values of NO concentration in post flame zone (a) and peak concentrations of NO (b), N₂O (c) and NO₂ (d) in NH₃/H₂/O₂/Ar flames vs ϕ at 4 atm.

shows experimental and numerical values of NO concentration in the post flame in the lean, stoichiometric, and rich NH₃/H₂/O₂/Ar flames, as well as NO peak concentration in the rich NH₃/H₂/O₂/Ar flame vs pressure.

At 1 atm, NO concentration in the post flame zone in the lean NH₃/H₂/O₂/Ar flame was $\sim 8 \times 10^{-3}$. Increasing pressure to 4 atm reduced NO concentration by about two times. A further pressure increase to 6 atm had a less significant effect. At 4 atm, NO concentration was $\sim 4 \times 10^{-3}$, and at 6 atm $\sim 3 \times 10^{-3}$ (Fig. 12a). According to experimental data for stoichiometric flames, NO concentration in the post flame zone at 1 atm was $\sim 5 \times 10^{-3}$. At 4 atm, NO concentration decreased to $\sim 2 \times 10^{-3}$. At 6 atm NO concentration was equal to $\sim 1.5 \times 10^{-3}$ (Fig. 12b). Figure 12c displays experimental and numerical values of NO concentration in the post flame zone in rich NH₃/H₂/O₂/Ar flames vs pressure. At 1 atm NO concentration in the post flame zone was $\sim 5 \times 10^{-4}$. A pressure increase to 4 atm resulted in the reduction of NO concentration to $\sim 1.0 \times 10^{-4}$. A further increase in pressure caused no noticeable change in NO concentration. Moreover, as noted, the detection limit of the mass spectrometric setup was the concentration of $\sim 10^{-5}$; it is not possible to show a noticeable change for smaller values. In Fig. 12d the experimental and numerical values of NO peak concentration are shown in the rich NH₃/H₂/O₂/Ar flames vs pressure. The experimental value of NO peak concentration at 1 atm was $\sim 2.2 \times 10^{-3}$. A pressure increase to 4 and 6 atm resulted in the reduction of NO concentration to 8.0×10^{-4} and 5.0×10^{-4} , respectively. In general, all Models qualitatively reproduce the trend of NO concentration (both peak and in the post-flame zone) reduction with the pressure increase. The largest discrepan-

cies are observed for Models 6, 7 and 8 in the case of lean and stoichiometric conditions and for Models 1, 2, 3, 6 and 7 for rich flames.

As previously noted, N₂O and NO₂ concentration in the post flame zone was negligible, therefore, only analysis of the pressure effect on the peak concentration of these species will be examined. Figure 13 shows experimental and numerical values of N₂O peak concentration in the lean, stoichiometric, and rich NH₃/H₂/O₂/Ar flames vs pressure.

At 1 atm, N₂O peak concentration in the lean NH₃/H₂/O₂/Ar flame was $\sim 1.8 \times 10^{-3}$. Increasing pressure to 4 and 6 atm reduced this value by about three times (Fig. 13a). According to Fig. 13b, peak concentration of N₂O in the stoichiometric flame at atmospheric pressure was $\sim 2.0 \times 10^{-3}$. At a pressure of 4 atm, this value became $\sim 2.5 \times 10^{-4}$. At 6 atm, the peak concentration of N₂O was equal to $\sim 2.0 \times 10^{-4}$. Figure 13c shows experimental and numerical values of N₂O peak concentration in the rich NH₃/H₂/O₂/Ar flame vs pressure. At 1 atm, the measured peak concentration of N₂O was $\sim 6.0 \times 10^{-4}$. At 4 atm, the peak concentration of N₂O decreased by about three times. At 6 atm, this value changed slightly. In general, all the tested models adequately reproduced the trend of peak concentration change with pressure.

Figure 14 presents experimental and numerical values of NO₂ peak concentration in the lean, stoichiometric, and rich NH₃/H₂/O₂/Ar flames vs pressure. Since NO₂ concentration profile was not measured at 1 atm, experimental values are given only for pressures of 4 and 6 atm.

The experimental data showed that the pressure increase resulted in a slight increase in NO₂ concentration. As it was indicated

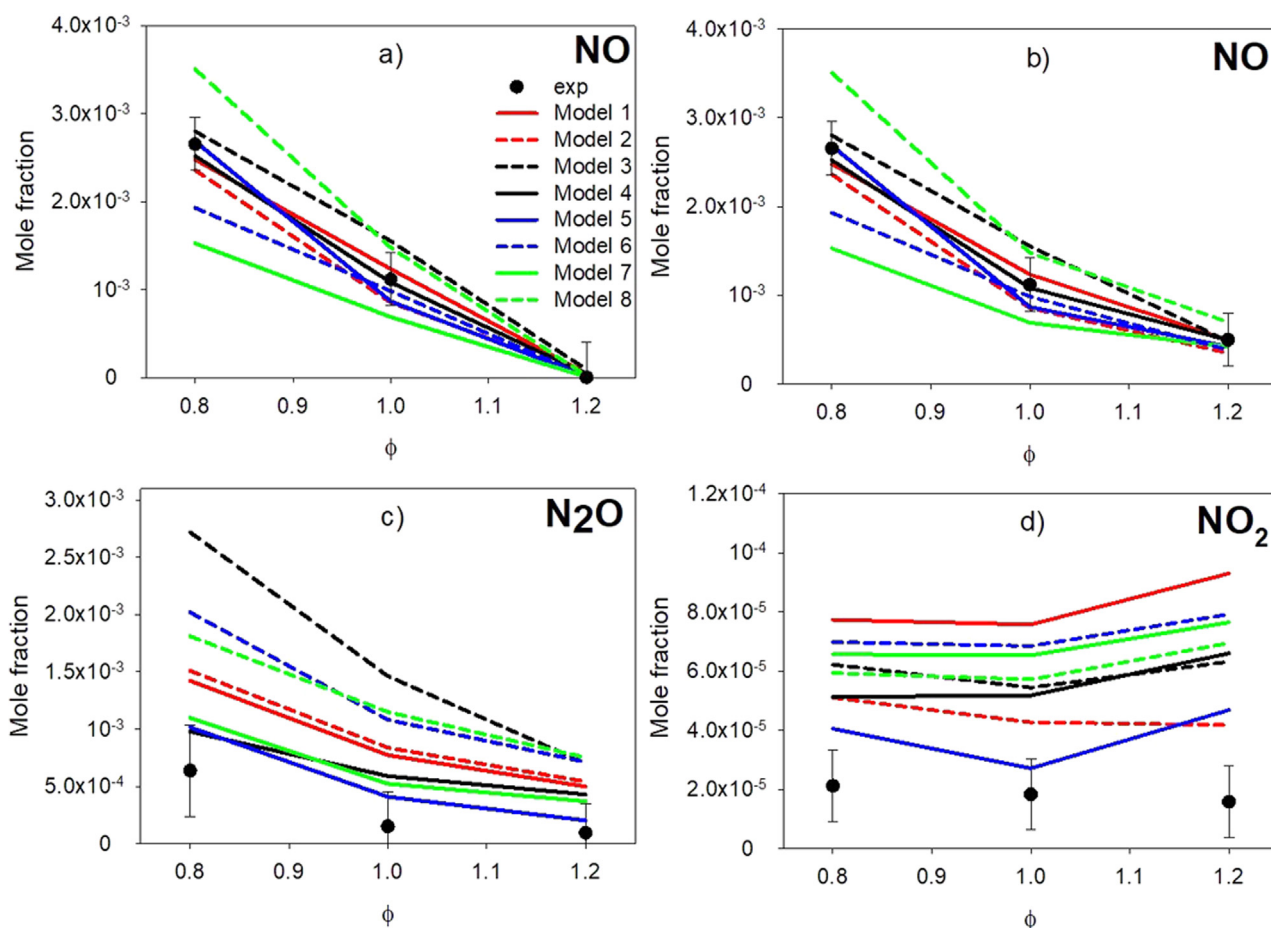


Fig. 11. Experimental and numerical values of NO concentration in post flame zone (a) and peak concentrations of NO (b), N_2O (c) and NO_2 (d) in $NH_3/H_2/O_2/Ar$ flames vs ϕ at 6 atm.

before, NO_2 maximum concentration is close to the detection limit of the experimental setup, thus, it is difficult to reach a conclusion about mechanisms predictive capability. However, it should be noted that Models used in this work predict different dependence of NO_2 concentration vs pressure.

The data presented revealed that to reduce the concentration of nitrogen oxides, it was necessary to consider the rich mixtures, since the transition to such mixtures makes possible reduced NO concentration (both peak and in the post flame zone) and peak concentration of N_2O . Although at $\phi=1.2$ NO_2 mole fraction slightly increased, even the peak concentration of this species was quite low. In the pressure effect, its increase also resulted in a decrease of NO concentration (both peak and post flame zone) and the peak concentration of N_2O . Like the equivalence ratio, the increase of pressure caused a slight increase in NO_2 peak concentration, but the absolute values of NO_2 peak concentration were very low.

This data showed that for mole fraction profiles of NO, the experimental and numerical data agreed well. In N_2O and NO_2 there were larger discrepancies between experimental and numerical data, especially for NO_2 . Therefore, the next section presents numerical calculations of the pathways of N_2O and NO_2 formation. Calculations were performed with Model 4 since this model demonstrated good predictive capability for the large set of experimental data and wide range of conditions [22,23,25]. To prove this statement, additional numerical modeling was performed, namely, for laminar flame speed and oxidation in a jet-stirred reactor. The experimental data for comparison were taken from the works

[23,25]. The results are shown in the Supplementary Material (Fig. 1S and Fig. 2S). In terms of laminar flame speed, the best agreement of experimental and modeling results is provided by Models 4 and 6. The mole fraction profile of NH_3 obtained in jet-stirred reactor indicates that Model 2 and 4 show the best predictive capability. Thus, Model 4 is able to correctly predict the large set of experimental data.

4.2.3. Numerical analysis of N_2O and NO_2 formation

The scheme illustrating the pathways of ammonia and hydrogen conversion to H_2O and N_2 is presented on the Fig. 3S. This scheme was obtained for stoichiometric flame at 4 atm since this pressure value as well as the value of equivalence ratio are intermediate for conditions studied in the present paper. Model 4 was used to determine the main reactions of fuels consumption. As can be seen, hydrogen is one of the products of ammonia conversion. This fact proves the idea that hydrogen which is suggested for improvement of ammonia combustion properties can be produced in the process of ammonia cracking. Hydrogen is mainly consumed in the reactions producing H_2O . Moreover, it should be noted that hydrogen produces radicals taking part in ammonia decomposition process. Ammonia decomposes with nitrogen oxides formation, which, in turn, take part in formation of H_2O and N_2 . As the above data indicated, the main problem of mechanisms predictive capability is associated with the NO_x sub-mechanism. Thus, to reveal the reactions needing the refinement the more detailed analysis of NO_x formation pathways from ammonia molecule is given below.

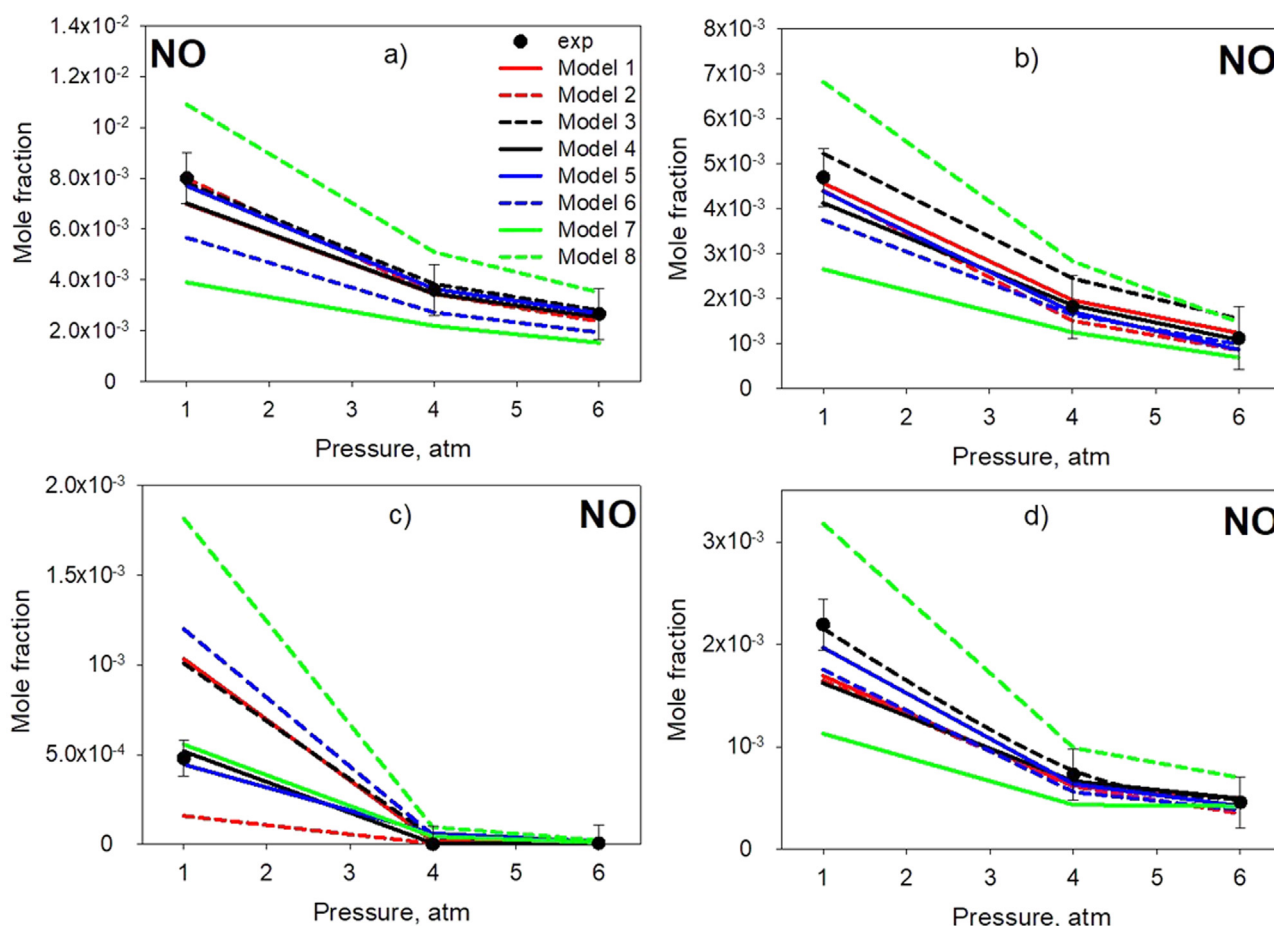


Fig. 12. Experimental and numerical values of NO concentration in post flame zone in lean (a), stoichiometric (b) and rich (c) $\text{NH}_3/\text{H}_2/\text{O}_2/\text{Ar}$ flames, and NO peak concentration in rich (d) $\text{NH}_3/\text{H}_2/\text{O}_2/\text{Ar}$ flame vs pressure.

Figure 15a shows a scheme of the main pathways of N_2O and NO_2 formation from the ammonia molecule at different equivalence ratios and the pressure of 6 atm. This pressure value was the highest in the set of experimental data presented and is more relevant to combustion chamber conditions in real engines and turbines. The analysis was performed with Model 4. Numbers near the arrows indicate the total contribution to the formation of a particular species in the $\text{NH}_3/\text{H}_2/\text{O}_2/\text{Ar}$ flame with the specific stoichiometry. Analysis of the reaction pathways was performed in the flame zone, corresponding to the maximum rate of ammonia consumption.

It can be seen from Fig. 15 that N_2O and NO_2 primarily formed in the following reactions: 1) H-abstraction from NH_3 by radicals OH, H and O with NH_2 formation; 2) H-abstraction from NH_2 by radicals OH, H and O with NH formation, or H substitution by O with HNO formation; 3) reactions of NH with OH and O_2 with HNO formation, or with O_2 and with NO formation; 4) NO formation in the reactions of HNO with OH, H, O, NH_2 , NH radicals, and with molecular oxygen; 5) N_2O formation in the reaction of NO with NH radical; 6) NO_2 forms from NO in its reactions with HO_2 and O radicals. Further the more detailed description the reactions leading to N_2O and NO_2 formation will be given.

The first step in the process of ammonia consumption is the formation of NH_2 -radical in reactions of NH_3 with H, O and OH. The total contribution of these reactions to the total flux of ammonia consumption is approximately 100%. The first step is almost independent on the fuel/oxidizer ratio. NH_2 in reactions with the same radicals (H, O and OH) yields NH-radical. In this case, there is some difference in the total contribution of these reactions when

the equivalence ratio changes, in particular, in rich conditions these reactions are less important (27.3%) compared to lean (34.1%) and stoichiometric ones (37.1%). This can be attributed to the fact that the total peak concentration of H, O and OH radicals is quite close in lean and stoichiometric flames (1.22×10^{-3} and 1.0×10^{-3} , respectively), while in the rich this value is around 0.27×10^{-3} (Fig. 9).

HNO forms in the reaction of NH_2 with O-radical. The contribution of these reactions, as expected, decreases in the transition from lean (11.4%) to rich (5.3%) conditions, since the concentration of atomic oxygen also decreases. For example, in the lean flame, the peak concentration of O-radicals is $\sim 8.0 \times 10^{-5}$, whereas in a rich flame it is only $\sim 2.0 \times 10^{-5}$. Another way of HNO formation includes the reactions of NH with OH-radical or molecular oxygen. The total contribution of these reactions decreases with the increase of equivalence ratio, since under rich conditions the concentration of both OH and available O_2 is lower. For example, the change in the equivalence ratio from 0.8 to 1.2 decreased the total contribution of these reactions approximately two times (from 27.2% to 12.5%).

NO forms from NH-radical in its reactions with O or O_2 . The total contribution of these reactions is lower in rich conditions compared to lean, since, as in the previous case, the concentration of O and O_2 is higher in lean flames. HNO mainly yields NO in reactions with radicals (H, O, OH, NH, NH_2) or molecular oxygen, the effect of equivalence ratio on the total contribution of these reactions is negligible. The total contribution of these reactions was more than 90% for all values of the equivalence ratio.

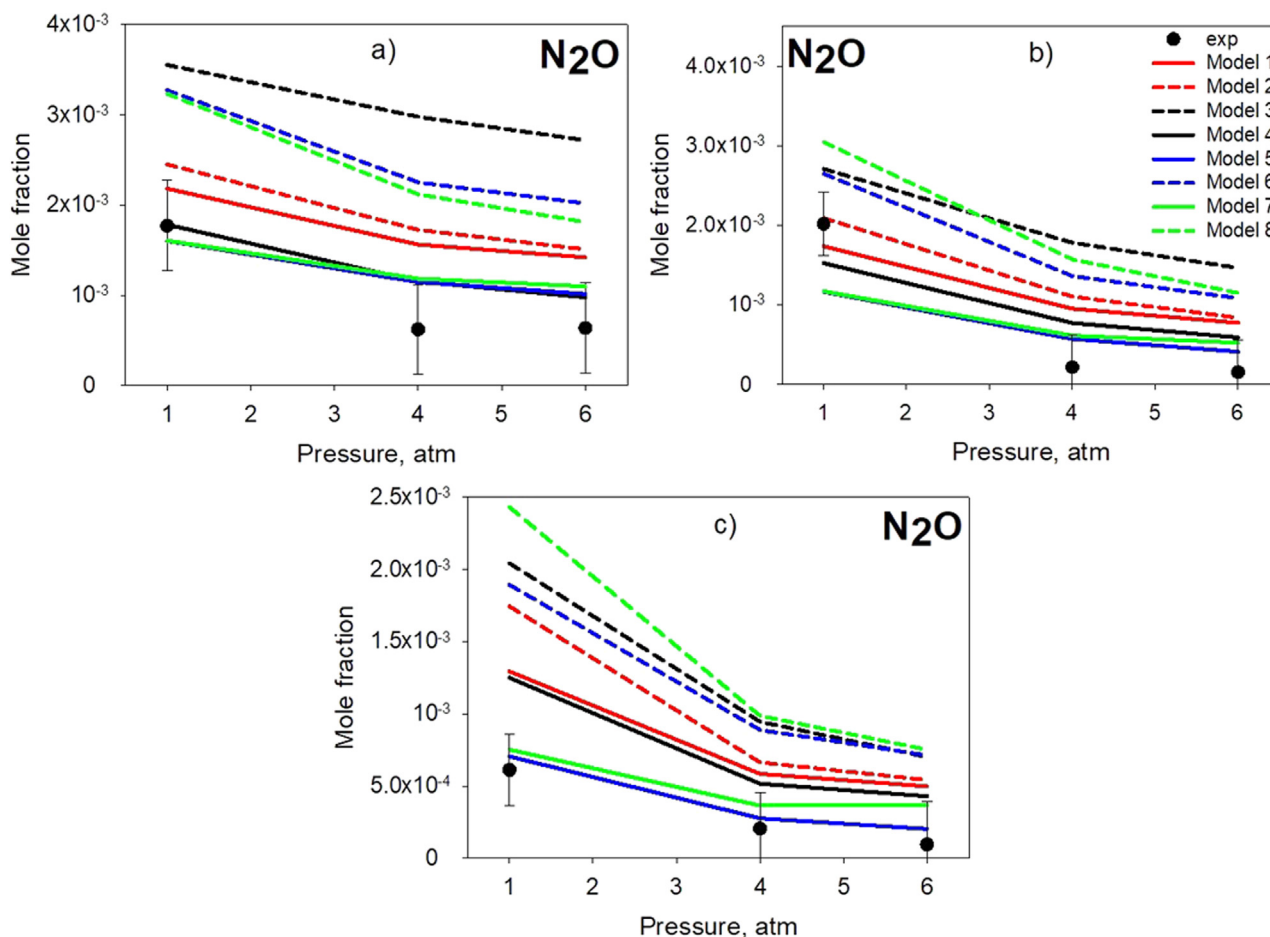


Fig. 13. Experimental and numerical values of N_2O peak concentration in lean (a), stoichiometric (b) and rich (c) $\text{NH}_3/\text{H}_2/\text{O}_2/\text{Ar}$ flames vs pressure.

N_2O mainly forms in the reaction of NO with NH , the contribution of this reaction changes slightly with the equivalence ratio increase. The contribution of the reaction of NO with NH to the total flux of NO consumption is 23.1% for lean conditions, 26.2% for stoichiometric and 21.9% for rich ones. The main pathways of NO_2 formation are reactions of NO with HO_2 or O . It should be noted that the total contribution of these reactions in lean conditions is almost two times higher compared to lean and stoichiometric ones. This can also be attributed to the higher concentration of O_2 and O in lean conditions. Namely, the total contribution of NO_2 formation reactions was 4.2% for the lean flame, 2.2% for the stoichiometric and 2.0% for the rich flame.

Figure 15b displays the scheme of the main pathways of N_2O and NO_2 formation from ammonia molecule in the stoichiometric $\text{NH}_3/\text{H}_2/\text{O}_2/\text{Ar}$ flames at pressures of 1, 4 and 6 atm. This value of the equivalence ratio was selected for analysis since it is intermediate in the experimental data. As in the case of Fig. 15a the analysis was performed with Model 4. The numbers near the arrows indicate total contribution to the formation of a particular species at a certain pressure. The reaction pathways were analyzed in the flame zone of the maximum rate of ammonia consumption. As shown, the pressure change did not cause the change of reaction pathways of N_2O and NO_2 formation.

As in the previous case, the first step in the process of ammonia consumption is H-abstraction with the formation of NH_2 radical. The total contribution of these reactions to the consumption of NH_3 is almost 100%. Thus, nearly complete ammonia conversion to NH_2 take place. Further, NH_2 yields either NH or HNO . NH forms in the reactions of NH_2 with the radicals H , O and

OH ; the total contribution of these reactions increases almost two times in the transition from atmospheric pressure to 4 atm and changes only slightly with a further increase in pressure to 6 atm. HNO mainly forms from NH_2 in the reaction of the latter with O -radical. At atmospheric pressure, the contribution of this channel to HNO formation is two times higher compared to 4 atm, whereas the further increase of pressure to 6 atm has only negligible effect.

NH is mainly consumed in the reactions with OH or O_2 , yielding HNO . Pressure increase results into the 2–3 times (from 6.9% at 1 atm to 18.7% at 4 atm and then to 19.6% at 6 atm) increase of the total contribution of these channels to NH consumption. NO forms in the reactions of NH with O_2 or O . On the contrary, at higher pressure (8.6% at 4 atm and 7.8% at 6 atm) these reactions become less important compared to the case of atmospheric pressure conditions (16.4%). The above scheme shows that HNO is mainly consumed in reactions with radicals H , O , OH , NH_2 , NH , as well as molecular oxygen, resulting in NO formation. The value of the total contribution of these reactions weakly depends on the pressure. The total contribution of these reactions was more than 90% for all pressure values. N_2O forms as the result of NO reaction with NH ; the channel of NO consumption becomes more important at higher pressures. For example, at 1 atm the contribution of this channel is 16%, while at 4 atm this value increases to 28.2%. The further pressure increase has very weak effect since at 6 atm the contribution of this reaction is 26.2%. The main pathways of NO_2 formation are the reactions of NO with HO_2 or O , whereas the total contribution of these reaction pathways changes only slightly with the pressure increase, namely, the contribution of NO_2 formation reactions to

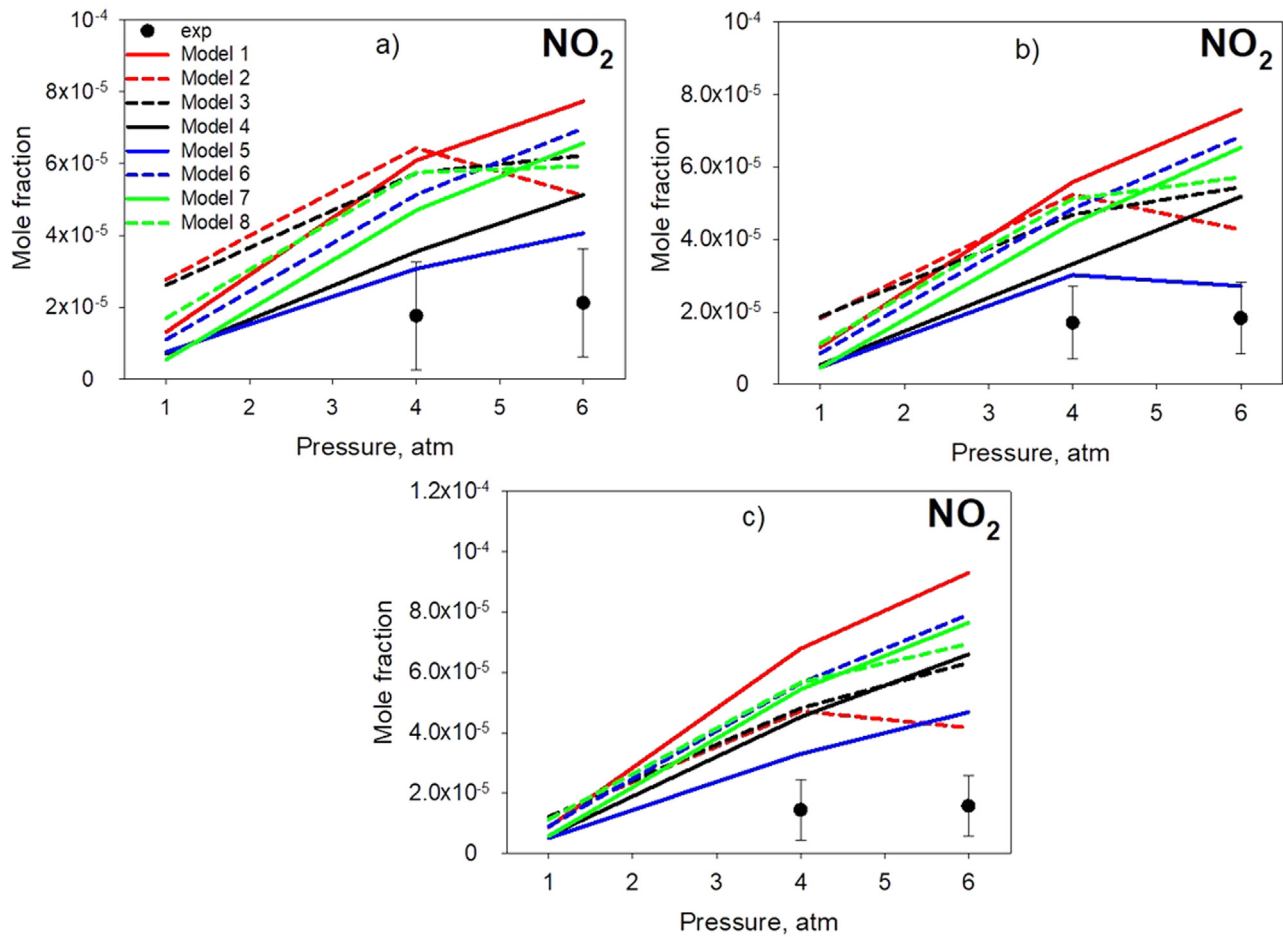


Fig. 14. Experimental and numerical values of NO_2 peak concentration in lean (a), stoichiometric (b) and rich (c) $\text{NH}_3/\text{H}_2/\text{O}_2/\text{Ar}$ flames vs pressure.

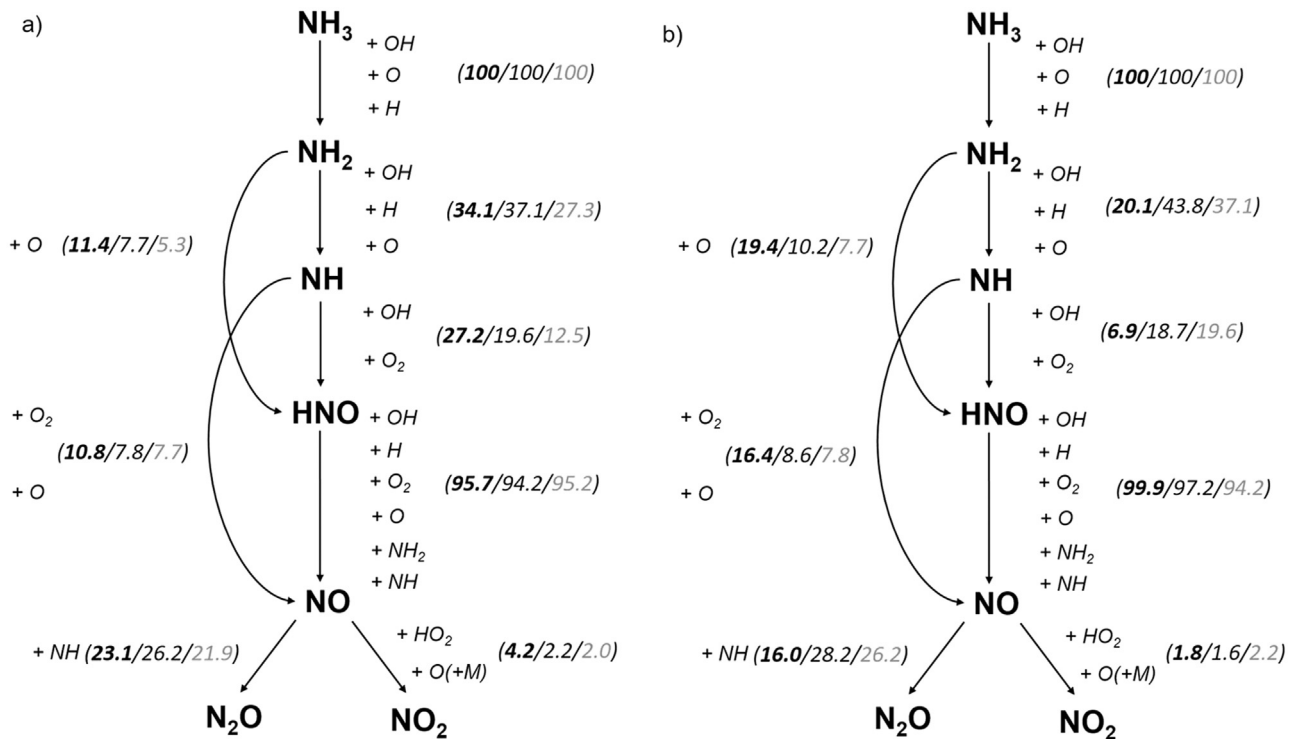


Fig. 15. Reaction pathways of N_2O and NO_2 formation from ammonia in lean/stoichiometric/rich $\text{NH}_3/\text{H}_2/\text{O}_2/\text{Ar}$ flames at 6 atm (a) and in the stoichiometric $\text{NH}_3/\text{H}_2/\text{O}_2/\text{Ar}$ flames at 1/4/6 atm (b).

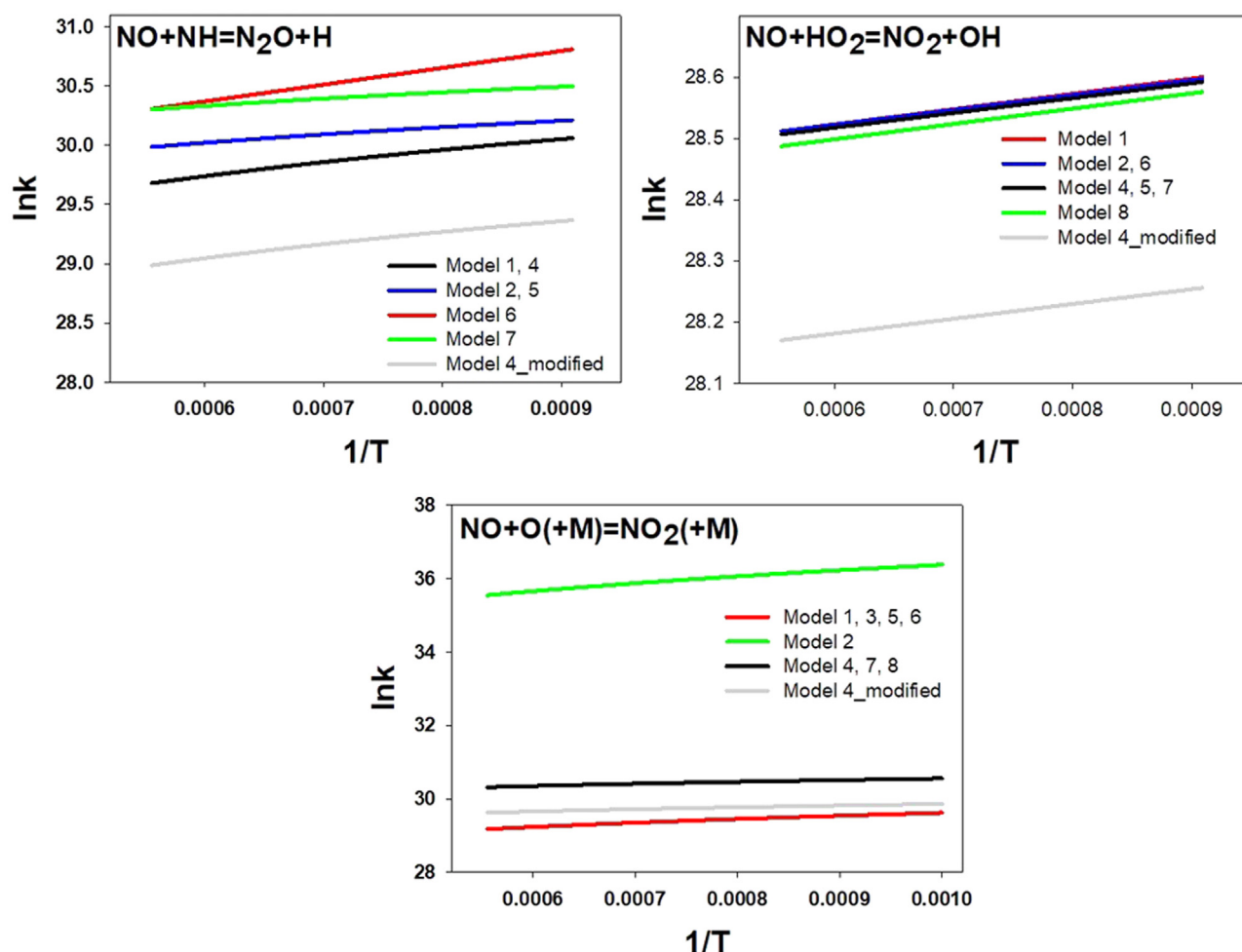


Fig. 16. Arrhenius plot for the reactions $\text{NO} + \text{NH} = \text{N}_2\text{O} + \text{H}$, $\text{NO} + \text{HO}_2 = \text{NO}_2 + \text{OH}$ and $\text{NO} + \text{O}(+M) = \text{NO}_2(+M)$.

the total consumption rate of NO was 1.8% at 1 atm, 1.6% at 4 atm, and 2.2% at 6 atm.

The data on the flame structure (Figs. 4,8) shows that the mole fraction profile of NO was described very well by Model 4. For N_2O concentration profiles there were larger discrepancies. For NO_2 , the agreement between experimental and numerical data was even worse. Analysis of N_2O and NO_2 formation revealed that these species formed from ammonia via NO. Therefore, to improve the agreement between experimental and numerical data for these species, it was necessary to pay attention to these reactions. Moreover, Fig. 15 shows that the contribution of N_2O formation reactions to NO consumption was not more than 30%, and the contribution of NO_2 formation reactions not more than 5% percent. Thus, it can be concluded that the change of the rate constants of the reactions of N_2O and NO_2 formation from NO, within the known accuracy of their estimation or measurement, should not have significantly affected the agreement between the experimental and numerical data for NO concentration profiles. An example of the numerical simulation using modified Model 4 is given in Supplementary Materials (Fig. 4S-16S). As previously noted, the main reaction of N_2O formation from NO was $\text{NO} + \text{NH} = \text{N}_2\text{O} + \text{H}$, and NO_2 formed mainly from NO in the reactions $\text{NO} + \text{HO}_2 = \text{NO}_2 + \text{OH}$ and $\text{NO} + \text{O}(+M) = \text{NO}_2(+M)$. In modified Model 4, the pre-exponential factors of the rate constants of the reactions $\text{NO} + \text{NH} = \text{N}_2\text{O} + \text{H}$ and $\text{NO} + \text{O}(+M) = \text{NO}_2(+M)$ were reduced two times, and in the case of $\text{NO} + \text{HO}_2 = \text{NO}_2 + \text{OH}$ reaction – 1.4 times, according to the un-

certainty limits provided in the work of Baulch [72]. However, as seen, this modification of the rate constants did not make it possible to achieve sufficient agreement for NO_2 concentration profiles, indicating that greater reduction of the rate constant of $\text{NO} + \text{HO}_2 = \text{NO}_2 + \text{OH}$ reaction was necessary. The example of numerical simulation with the Model 4, in which the pre-exponential factor of the rate constant of $\text{NO} + \text{HO}_2 = \text{NO}_2 + \text{OH}$ was reduced two times, is provided in Supplementary Material (Fig. 17S-28S). This modification of the mechanism provided better agreement between experimental and modeling results for NO_2 concentration profile.

In this work, the rate constants of the key reactions of N_2O and NO_2 formation from NO, which are present in all tested mechanisms, were analyzed as well. Table 4 provides the Arrhenius parameters of the rate constants of these reactions $\text{NO} + \text{NH} = \text{N}_2\text{O} + \text{H}$, $\text{NO} + \text{HO}_2 = \text{NO}_2 + \text{OH}$, $\text{NO} + \text{O}(+M) = \text{NO}_2(+M)$.

Figure 16 illustrates the temperature dependence of these rate constants in Arrhenius coordinates. This figure also shows the Arrhenius plots for the rate constants, which were obtained with the manual reduction of the pre-exponential factors, as it was mentioned in the previous paragraph.

As can be seen, for the reactions $\text{NO} + \text{NH} = \text{N}_2\text{O} + \text{H}$ and $\text{NO} + \text{HO}_2 = \text{NO}_2 + \text{OH}$ the modified rate constants are much lower compared to those taken from other sources (Models 1-8). Thus, the adoption of any of these rate constants will not result into the improvement of agreement between experimental and

Table 4Arrhenius parameters of the rate constants for the reaction $\text{NO} + \text{NH} = \text{N}_2\text{O} + \text{H}$, $\text{NO} + \text{HO}_2 = \text{NO}_2 + \text{OH}$, $\text{NO} + \text{O}(+M) = \text{NO}_2(+M)$.

Model №	Rate constants parameters								
	$\text{NO} + \text{NH} = \text{N}_2\text{O} + \text{H}$			$\text{NO} + \text{HO}_2 = \text{NO}_2 + \text{OH}$			$\text{NO} + \text{O}(+M) = \text{NO}_2(+M)$		
	A	n	E	A	n	E	A	n	E
Model 1	2.7E+15	−0.78	20	2.10E+12	0	−497	1.30E+15	−0.75	0.0
Model 2	2.90E+14	−0.4	0.0	2.11E+12	0	−480	1.06E+20	−1.41	0.0
Model 3 (reverse)	6.70E+22	−2.16	37,155	1.81E+13	0	6676	1.30E+15	−0.75	0.0
Model 4	2.70E+15	−0.78	20	2.10E+12	0	−480	2.95E+14	−0.4	0.0
Model 5	2.90E+14	−0.4	0.0	2.10E+12	0	−480	1.30E+15	−0.75	0.0
	−2.20E+13	−0.23	0.0						
Model 6	5.33E+12	0.026	−2893.9	2.11E+12	0	−480	1.30E+15	−0.75	0.0
Model 7	2.90E+14	−0.4	0.0	2.10E+12	0	−480	2.95E+14	−0.4	0.0
Model 8 (reverse)	7.34E+20	−1.55	36,940	2.05E+12	0	−497	2.95E+14	−0.4	0.0

Units: A - mole-cm-sec-K; E - cal/mole.

numerical data on N_2O and NO_2 concentration profiles. For $\text{NO} + \text{O}(+M) = \text{NO}_2(+M)$, in the contrary, there are rate constants which are even lower compared to the modified one. For this reason, the test simulation using Model 4 with the adopted rate constant of $\text{NO} + \text{O}(+M) = \text{NO}_2(+M)$ reaction (from Models 1,3,5,6) was performed. The conditions were similar to those for stoichiometric flame at 4 atm. For brevity, only results for NO, N_2O and NO_2 are provided in Supplementary Materials. (Fig. 29S). These graphs indicate that there is a slight effect on NO concentration in the post-flame zone. For N_2O , as expected, there is almost no effect. For NO_2 the change of the peak concentration when using Model 4 with the adopted rate constant (from Models 1,3,5,6) is not more than 2%. This modification is definitely not enough to improve the agreement of experiment and modeling for N_2O and NO_2 . Thus, the reaction scheme for nitrogen containing species in Model 4 needs further refinement.

5. Conclusions

New experimental data on the $\text{NH}_3/\text{H}_2/\text{O}_2/\text{Ar}$ flames structure at 4 and 6 atm have been presented in this work. Eight published detailed chemical-kinetic models were tested against the experimental data. Experimental data and numerical simulation showed that the main nitrogen containing species present in the post flame zone were N_2 and NO, whereas N_2O and NO_2 concentration in the post flame zone was very low. It was also revealed that transition to rich conditions ($\varphi=1.2$) resulted in the reduction of NO concentration in the post flame zone, as well as in the reduction of peak concentrations of NO and N_2O in the reaction zone. Increasing the pressure from 1 to 6 atm also decreased the concentration of NO in the post flame, and the peak concentration of NO and N_2O . For the trend of NO concentration versus pressure, there was satisfactory agreement between experimental and numerical data. There were larger discrepancies for N_2O between experimental and modeling results. The greatest discrepancies between experimental and numerical data were observed with NO_2 . Numerical analysis of N_2O and NO_2 formation pathways revealed that these species mainly formed from NO. The contribution of the reactions of N_2O formation to the total NO consumption rate was not more than 30%, and the reactions of NO_2 formation contributed not more than 5%. Therefore, to provide better agreement for N_2O and NO_2 concentration profiles, it is necessary to refine the rate constants of reactions involving NO. The data obtained on NO_2 concentration profiles indicate that more studies of NO_2 -chemistry are required.

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Associated content

Supplementary Materials.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.combustflame.2022.112419](https://doi.org/10.1016/j.combustflame.2022.112419).

References

- [1] J. Chen, Y. Fei, Z. Wan, The relationship between the development of global maritime fleets and GHG emission from shipping, *J. Environ. Manage.* 242 (2019) 31–39.
- [2] Y. Kojima, Hydrogen storage materials for hydrogen and energy carriers, *Int. J. Hydrog. Energy* 44 (2019) 18179–18192.
- [3] N. Morlanés, S.P. Katikaneni, S.N. Paglieri, A. Harale, B. Solami, S.M. Sarathy, J.A. Gascon, Technological roadmap to the ammonia energy economy: current state and missing technologies, *Chem. Eng. J.* 408 (2021) 127310.
- [4] R. Service, Liquid sunshine, *Science* 361 (2018) 120–123.
- [5] K. Takizawa, A. Takahashi, K. Tokuhashi, S. Kondo, A. Sekiya, Burning velocity measurements of nitrogen-containing compounds, *J. Hazard. Mater.* 155 (2008) 144–152.
- [6] A. Hayakawa, T. Goto, R. Mimoto, Y. Arakawa, T. Kudo, H. Kobayashi, Laminar burning velocity and Markstein length of ammonia/air premixed flames at various pressures, *Fuel* 159 (2015) 98–106.
- [7] J.H. Lee, J.H. Kim, J.H. Park, O.C. Kwon, Studies on properties of laminar premixed hydrogen-added ammonia/air flames for hydrogen production, *Int. J. Hydrog. Energy* 35 (2010) 1054–1064.
- [8] A. Ichikawa, A. Hayakawa, Y. Kitagawa, K.D.K.A. Somaratne, T. Kudo, H. Kobayashi, Laminar burning velocity and Markstein length of ammonia/hydrogen/air premixed flames at elevated pressures, *Int. J. Hydrog. Energy* 40 (2015) 9570–9578.
- [9] R.C. da Rocha, M. Costa, X.-S. Bai, Chemical kinetic modelling of ammonia/hydrogen/air ignition, premixed flame propagation and NO emission, *Fuel* 246 (2019) 24–33.
- [10] X. Han, Z. Wang, M. Costa, Z. Sun, Y. He, K. Cen, Experimental and kinetic modeling study of laminar burning velocities of NH_3/Air , $\text{NH}_3/\text{H}_2/\text{Air}$, $\text{NH}_3/\text{CO}/\text{air}$ and $\text{NH}_3/\text{CH}_4/\text{air}$ premixed flames, *Combust. Flame* 206 (2019) 214–226.
- [11] J. Li, H. Huang, N. Kobayashi, Z. He, Y. Nagai, Study on using hydrogen and ammonia as fuels: combustion characteristics and NOx formation, *Int. J. Energy Res* 38 (2014) 1214–1223.
- [12] J.-B. Lee, S.-I. Lee, O.C. Kwon, Effects of ammonia substitution on hydrogen/air flame propagation and emissions, *Int. J. Hydrog. Energy* 35 (2010) 11332–11341.
- [13] S. Wang, Z. Wang, A.M. Elbaz, X. Han, Y. He, M. Costa, A.A. Konnov, W.L. Roberts, Experimental study and kinetic analysis of the laminar burning velocity of $\text{NH}_3/\text{syngas}/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$ and $\text{NH}_3/\text{H}_2/\text{air}$ premixed flames at elevated pressures, *Combust. Flame* 221 (2020) 270–287.
- [14] C. Lhuillier, P. Brequigny, N. Lamoureux, F. Contino, C. Mounaïm-Rouselle, Experimental investigation on laminar burning velocities of ammonia/hydrogen/air mixtures at elevated temperatures, *Fuel* 263 (2020) 116653.

- [15] B. Mei, J. Zhang, X. Shi, Z. Xi, Y. Li, Enhancement of ammonia combustion with partial fuel cracking strategy: laminar flame propagation and kinetic modeling investigation of $\text{NH}_3/\text{H}_2/\text{N}_2/\text{air}$ mixtures up to 10 atm, *Combust. Flame* 231 (2021) 111472.
- [16] K.P. Shrestha, C. Lhuillier, A.A. Barbosa, P. Brequigny, F. Contino, C. Mounaim-Rousselle, L. Seidel, F. Mauss, An experimental and modeling study of ammonia with enriched oxygen content and ammonia/hydrogen laminar flame speed at elevated pressure and temperature, *Proc. Combust. Inst.* 38 (2021), pp. 2163–2174.
- [17] G.J. Gotama, A. Hayakawa, E.C. Okafor, R. Kanoshima, M. Hayashi, T. Kudo, H. Kobayashi, Measurement of the laminar burning velocity and kinetics study of the importance of the hydrogen recovery mechanism of ammonia/hydrogen/air premixed flames, *Combust. Flame* 236 (2022) 111753.
- [18] B. Shu, S.K. Vallabhuni, X. He, G. Issayev, K. Moshhammer, A. Farooq, R.X. Fernandes, A shock tube and modeling study on the autoignition properties of ammonia at intermediate temperatures, *Proc. Combust. Inst.* 37 (2019), pp. 205–211.
- [19] L. Dai, S. Gersen, P. Glarborg, H. Levinsky, A. Mokhov, Experimental and numerical analysis of the autoignition behavior of NH_3 and NH_3/H_2 mixtures at high pressure, *Combust. Flame* 215 (2020) 134–144.
- [20] O. Mathieu, E.L. Petersen, Experimental and modeling study on the high-temperature oxidation of ammonia and related NOx chemistry, *Combust. Flame* 162 (2015) 554–570.
- [21] P. Sabia, M.V. Manna, R. Ragucci, M. de Joannon, Mutual inhibition effect of hydrogen and ammonia in oxidation processes and the role of ammonia as “strong” collider in third-molecular reactions, *Int. J. Hydrog. Energy* 45 (2020) 32113–32127.
- [22] X. Zhang, S.P. Moosakutty, R.P. Rajan, M. Younes, S.M. Sarathy, Combustion chemistry of ammonia/hydrogen mixtures: jet-stirred reactor measurements and comprehensive kinetic modeling, *Combust. Flame* 234 (2021) 111653.
- [23] K.N. Osipova, X. Zhang, S.M. Sarathy, O.P. Korobeinichev, A.G. Shmakov, Ammonia and ammonia/hydrogen blends oxidation in a jet-stirred reactor: experimental and numerical study, *Fuel* 310 (2022) 122202.
- [24] C. Duynslaegher, H. Jeanmart, J. Vandooren, Flame structure studies of premixed ammonia/hydrogen/oxygen/argon flames: experimental and numerical investigation, *Proc. Combust. Inst.* 32 (2009), pp. 1277–1284.
- [25] K.N. Osipova, O.P. Korobeinichev, A.G. Shmakov, Chemical structure and laminar burning velocity of atmospheric pressure premixed ammonia/hydrogen flames, *Int. J. Hydrog. Energy* 46 (2021) 39942–39954.
- [26] J.A. Miller, C.T. Bowman, Mechanism and modeling of nitrogen chemistry in combustion, *Prog. Energy Combust. Sci.* 15 (1989) 287–338.
- [27] W. Tsang, R.F. Hampson, Chemical kinetic data base for combustion chemistry. Part I. Methane and related compounds, *J. Phys. Chem. Ref. Data* 15 (1986) 1087–1279.
- [28] R.P. Lindstedt, F.C. Lockwood, M.A. Selim, Detailed kinetic modelling of chemistry and temperature effects on ammonia oxidation, *Combust. Sci. Technol.* 99 (1994) 253–276.
- [29] Y. Song, H. Hashemi, J.M. Christensen, C. Zou, P. Marshall, P. Glarborg, Ammonia oxidation at high pressure and intermediate temperatures, *Fuel* 181 (2016) 358–365.
- [30] P. Glarborg, J.A. Miller, B. Ruscic, S.J. Klippenstein, Modeling nitrogen chemistry in combustion, *Prog. Energy Combust. Sci.* 67 (2018) 31–68.
- [31] K.P. Shrestha, L. Seidel, T. Zeuch, F. Mauss, Detailed kinetic mechanism for the oxidation of ammonia including the formation and reduction of nitrogen oxides, *Energy Fuels* 32 (2018) 10202–10217.
- [32] J. Otomo, M. Koshi, T. Mitsumori, H. Iwasaki, K. Yamada, Chemical kinetic modeling of ammonia oxidation with improved reaction mechanism for ammonia/air and ammonia/hydrogen/air combustion, *Int. J. Hydrog. Energy* 43 (2018) 3004–3014.
- [33] F.H.V. Coppens, J. De Ruyck, A.A. Konnov, The effects of composition on burning velocity and nitric oxide formation in laminar premixed flames of $\text{CH}_4 + \text{H}_2 + \text{O}_2 + \text{N}_2$, *Combust. Flame* 149 (2007) 409–417.
- [34] T. Mendiara, P. Glarborg, Ammonia chemistry in oxy-fuel combustion of methane, *Combust. Flame* 156 (2009) 1937–1949.
- [35] S.J. Klippenstein, L.B. Harding, P. Glarborg, J.A. Miller, The role of NNH in NO formation and control, *Combust. Flame* 158 (2011) 774–789.
- [36] A.G. Shmakov, O.P. Korobeinichev, I.V. Rybitskaya, A.A. Chernov, D.A. Knyazkov, T.A. Bolshova, Konnov A.A., Formation and consumption of NO in $\text{H}_2 + \text{O}_2 + \text{N}_2$ flames doped with NO or NH_3 at atmospheric pressure, *Combust. Flame* 157 (2010) 556–565.
- [37] C. Brackmann, V.A. Alekseev, B. Zhou, E. Nordström, P.-E. Bengtsson, Z. Li, M. Aldén, A.A. Konnov, Structure of premixed ammonia + air flames at atmospheric pressure: laser diagnostics and kinetic modeling, *Combust. Flame* 163 (2016) 370–381.
- [38] E.C. Okafor, Y. Naito, S. Colson, A. Ichikawa, T. Kudo, A. Hayakawa, H. Kobayashi, Experimental and numerical study of the laminar burning velocity of $\text{CH}_4\text{--NH}_3\text{--air}$ premixed flames, *Combust. Flame* 187 (2018) 185–198.
- [39] G.P. Smith, D.M. Golden, M. Frenklach, N.W. Moriarty, B. Eiteneer, M. Goldenberg, C.T. Bowman, R.K. Hanson, S. Song, W.C. Gardiner, Jr., V.V. Lissianski, Z. Qin, <http://combustion.berkeley.edu/gri-mech/>.
- [40] Z. Tian, Y. Li, L. Zhang, P. Glarborg, F. Qi, An experimental and kinetic modeling study of premixed $\text{NH}_3/\text{CH}_4/\text{O}_2/\text{Ar}$ flames at low pressure, *Combust. Flame* 156 (2009) 1413–1426.
- [41] B. Mei, X. Zhang, S. Ma, M. Cui, H. Guo, Z. Cao, Y. Li, Experimental and kinetic modeling investigation on the laminar flame propagation of ammonia under oxygen enrichment and elevated pressure conditions, *Combust. Flame* 210 (2019) 236–246.
- [42] H. Hashemi, J.M. Christensen, S. Gersen, P. Glarborg, Hydrogen oxidation at high pressure and intermediate temperatures: experiments and kinetic modeling, *Proc. Combust. Inst.* 35 (2015), pp. 553–560.
- [43] A. Stagni, C. Cavallotti, S. Arunthanayothin, Y. Song, O. Herbinet, F. Battin-Leclerc, T. Faravelli, An experimental, theoretical and kinetic-modeling study of the gas-phase oxidation of ammonia, *React. Chem. Eng.* 5 (2020) 696–711.
- [44] W.K. Metcalfe, S.M. Burke, S.S. Ahmed, H.J. Curran, A hierarchical and comparative kinetic modeling study of $\text{C}_1 - \text{C}_2$ hydrocarbon and oxygenated fuels, *Int. J. Chem. Kinet.* 45 (2013) 638–675.
- [45] Y. Song, L. Marrodán, N. Vin, O. Herbinet, E. Assaf, C. Fittschen, A. Stagni, T. Faravelli, M.U. Alzueta, F. Battin-Leclerc, The sensitizing effects of NO_2 and NO on methane low temperature oxidation in a jet stirred reactor, *Proc. Combust. Inst.* 37 (2019), pp. 667–675.
- [46] X. Chen, M.E. Fuller, C.F. Goldsmith, Decomposition kinetics for HONO and HNO_2 , *React. Chem. Eng.* 4 (2019) 323–333.
- [47] A. Burcat, B. Ruscic, Third Millennium Ideal Gas and Condensed Phase Thermochemical Database for Combustion with Updates from Active Thermochemical Tables B., 2005 vol. ANL-05/20.
- [48] A. Bertolino, M. Fürst, A. Stagni, A. Frassoldati, M. Pelucchi, C. Cavallotti, T. Faravelli, A. Parente, An evolutionary, data-driven approach for mechanism optimization: theory and application to ammonia combustion, *Combust. Flame* 229 (2021) 111366.
- [49] R. Li, A.A. Konnov, G. He, F. Qin, D. Zhang, Chemical mechanism development and reduction for combustion of $\text{NH}_3/\text{H}_2/\text{CH}_4$ mixtures, *Fuel* 257 (2019) 116059.
- [50] NUIG Combustion Chemistry Center. <http://c3.nuigalway.ie/combustionchemistrycentre/mechanismdownloads/>.
- [51] B. Mei, S. Ma, Y. Zhang, X. Zhang, W. Li, Y. Li, Exploration on laminar flame propagation of ammonia and syngas mixtures up to 10 atm, *Combust. Flame* 220 (2020) 368–377.
- [52] J.A. Miller, C.F. Melius, The reactions of imidogen with nitric oxide and molecular oxygen, Twenty-Fourth Symp. Combust. 24 (1992) 719–726.
- [53] R. Sumathi, D. Sengupta, M.T. Nguyen, Theoretical study of the $\text{H}_2 + \text{NO}$ and related reactions of $[\text{H}_2\text{NO}]$ isomers, *J. Phys. Chem. A* 102 (1998) 3175–3183.
- [54] X. Han, M.L. Lavadera, A.A. Konnov, An experimental and kinetic modeling study on the laminar burning velocity of $\text{NH}_3 + \text{N}_2\text{O} + \text{air}$ flames, *Combust. Flame* 228 (2021) 13–28.
- [55] A.A. Konnov, J. De Ruyck, Kinetic modeling of the decomposition and flames of hydrazine, *Combust. Flame* 124 (2001) 106–126.
- [56] A.A. Konnov, G. Colson, J. De Ruyck, The new route forming NO via NNH, *Combust. Flame* 121 (2000) 548–550.
- [57] A.A. Konnov, J. De Ruyck, Temperature-dependent rate constant for the reaction $\text{NNH} + \text{O} \rightarrow \text{NH} + \text{NO}$, *Combust. Flame* 125 (2001) 1258–1264.
- [58] A.A. Konnov, G. Colson, J. De Ruyck, NO formation rates for hydrogen combustion in stirred reactors, *Fuel* 80 (2001) 49–65.
- [59] A.M. Dmitriev, K.N. Osipova, D.A. Knyazkov, I.E. Gerasimov, A.G. Shmakov, O.P. Korobeinichev, Comparative analysis of the chemical structure of ethyl butanoate and methyl pentanoate flames, *Combust. Explos. Shock Waves* 54 (2018) 125–135.
- [60] I.E. Gerasimov, D.A. Knyazkov, S.A. Yakimov, T.A. Bolshova, A.G. Shmakov, O.P. Korobeinichev, Structure of atmospheric-pressure fuel-rich premixed ethylene flame with and without ethanol, *Combust. Flame* 159 (2012) 1840–1850.
- [61] O.P. Korobeinichev, V.M. Shvartsberg, A.G. Shmakov, D.A. Knyazkov, I.V. Rybitskaya, Inhibition of atmospheric lean and rich $\text{CH}_4/\text{O}_2/\text{Ar}$ flames by phosphorus-containing compound, *Proc. Combust. Inst.* 31 (2007), pp. 2741–2748.
- [62] J.P. Botha, D.B. Spalding, A.C. Egerton, The laminar flame speed of propane/air mixtures with heat extraction from the flame, *Proc. R. Soc. Lond. Ser. Math. Phys. Sci.* 225 (1954), pp. 71–96.
- [63] A.M. Dmitriev, D.A. Knyazkov, T.A. Bolshova, A.G. Tereshchenko, A.A. Paletsky, A.G. Shmakov, O.P. Korobeinichev, Structure of $\text{CH}_4/\text{O}_2/\text{Ar}$ flames at elevated pressures studied by flame sampling molecular beam mass spectrometry and numerical simulation, *Combust. Flame* 162 (2015) 3946–3959.
- [64] D.A. Knyazkov, T.A. Bolshova, A.M. Dmitriev, A.G. Shmakov, O.P. Korobeinichev, Experimental and numerical investigation of the chemical reaction kinetics in H_2/CO syngas flame at a pressure of 1–10 atm, *Combust. Explos. Shock Waves* 53 (2017) 388–397.
- [65] D.A. Knyazkov, A.M. Dmitriev, T.A. Bolshova, V.M. Shvartsberg, A.G. Shmakov, O.P. Korobeinichev, Structure of premixed $\text{H}_2/\text{O}_2/\text{Ar}$ flames at 1–5 atm studied by molecular beam mass spectrometry and numerical simulation, *Proc. Combust. Inst.* 36 (2017), pp. 1233–1240.
- [66] D.A. Knyazkov, A.M. Dmitriev, V.M. Shvartsberg, K.N. Osipova, A.G. Shmakov, O.P. Korobeinichev, Study of the chemical structure of laminar premixed $\text{H}_2/\text{CH}_4/\text{C}_3\text{H}_8/\text{O}_2/\text{Ar}$ flames at 1–5 atm, *Energy Fuels* 31 (2017) 11377–11390.
- [67] O.P. Korobeinichev, A.G. Shmakov, I.V. Rybitskaya, T.A. Bolshova, A.A. Chernov, D.A. Knyazkov, A.A. Konnov, Kinetics and mechanism of chemical reactions in the $\text{H}_2/\text{O}_2/\text{N}_2$ flame at atmospheric pressure, *Kinet. Catal.* 50 (2009) 156–161.
- [68] <https://www.nist.gov/pml/electron-impact-cross-sections-ionization-and-excitation-database>.
- [69] W.L. Fitch, A.D. Sauter, Calculation of relative electron impact total ionization cross sections for organic molecules, *Anal. Chem.* 55 (1983) 832–835.
- [70] A. Lamprecht, B. Atakan, K. Kohse-Hö, Fuel-rich propene and acetylene flames: a comparison of their flame chemistries, *Combust. Flame* 122 (2000) 483–491.

- [71] O.P. Korobeinichev, S.B. Ilyin, V.V. Mokrushin, A.G. Shmakov, Destruction chemistry of dimethyl methylphosphonate in H_2/O_2 /Ar flame studied by molecular beam mass spectrometry, *Combust. Sci. Technol.* 116–117 (1996) 51–67.
- [72] D.L. Baulch, C.T. Bowman, C.J. Cobos, R.A. Cox, Th. Just, J.A. Kerr, M.J. Pilling, D. Stocker, J. Troe, W. Tsang, R.W. Walker, J. Warnatz, Evaluated kinetic data for combustion modeling: supplement II, *J. Phys. Chem. Ref. Data* 34 (2005) 757–1397.